Multimodal-Interactive GCN-Guided Space–Frequency Implicit Fusion Network for Semantic Segmentation of Optical and SAR Images

Bingxia Xin, Hongqing Li, Chunyu Zhu, Rongyuan Dai, and Qiong Wu

***Abstract*—Multisource remote sensing images capture complementary information of ground objects and provide valuable inputs for semantic segmentation, and different sensors may produce modality inconsistencies and interference noise. Optical and synthetic aperture radar (SAR) segmentation is challenging for three main reasons. First, geometric distortions introduce scale inconsistencies. Second, speckle noise degrades signal quality. Third, subtle interclass differences lead to misalignment between modalities. Collectively, these factors aggravate cross-modal contextual inconsistency and hinder accurate multimodal segmentation. We propose MSFNet, a graph-guided spatial-frequency implicit fusion network that unifies graph priors with continuous implicit representations. A dual-path encoder processes optical images with a convolutional neural network (CNN) and SAR with a dynamic adaptive graph convolutional network (DA-GCN) to capture topological/scattering structures. Fusion is performed by graph-guided spatial implicit sampling (GSIS) for geometry-aware alignment in the continuous coordinate space and graph-guided Fourier implicit sampling (GFIS) for frequency-aware detail compensation and global context modulation. A multiscale decoder yields fine-grained boundaries. On WHU-OPT-SAR, DFC23, YYX-OPT-SAR, and DFC2025, MSFNet achieves state-of-the-art performance, with consistent gains in boundary $F1$-score (Boundary-$F1$) and boundary intersection-over-union (bIoU), as well as robustness to misregistration and speckle. These results demonstrate that structure-aware graph modeling and spatial-frequency implicit fusion jointly improve accuracy and robustness for multimodal remote-sensing segmentation. The code will be shared on GitHub soon.**

***Index Terms*—Graph convolution, implicit neural representation (INR), multimodal image fusion, remote sensing semantic segmentation, spatial-frequency dual-domain implicit guidance.**

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Bingxia Xin, Rongyuan Dai, and Qiong Wu are with the College of Geo-Exploration Science and Technology, Jilin University, Changchun 130026, China (e-mail: xinbx23@mails.jlu.edu.cn; dairongyuan25@mails.jlu.edu.cn; wuqiong@jlu.edu.cn).

Hongqing Li is with the College of Geo-Exploration Science and Technology, Jilin University, Changchun 130026, China, and also with the College of Natural Resource Science and Technology, Xinjiang University of Technology, Hotan, Xinjiang 848000, China (e-mail: Lihq@jlu.edu.cn).

Chunyu Zhu is with Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China (e-mail: zhuchunyu@xidian.edu.cn).

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I. Introduction

SEMANTIC segmentation of remote sensing images on the pixel-level has been widely applied in ecological monitoring and urban planning [1], [2], [3]. Big data development of remote sensing has produced an increasing number of optical and synthetic aperture radar (SAR) images [4]. Optical images with rich spectral information are susceptible to weather and illumination changes, resulting in unstable segmentation performance. SAR images, with the advantage of all-weather active imaging, suffer from the noise that limits the expression of detailed information. SAR images exhibit distinct backscattering differences near homogeneous land covers (e.g., farmland and water bodies), which facilitates accurate boundary delineation [Fig. 1(a)]. Similarly, optical images exhibit more salient texture differences in transitional areas between buildings and farmland [Fig. 1(b)]. The fusion of optical and SAR images (OSIs) can not only alleviate class ambiguity to enhance feature completeness but also significantly improve semantic segmentation accuracy and robustness in complex scenes [5], [6].

Optical and SAR semantic segmentation generally includes two stages [7]. In the first stage, OSI features are extracted through a dual-branch network. In the second stage, feature fusion is performed based on modality differences and task requirements [2], [8]. Traditional methods extract handcrafted feature vectors using multiscale filters (Gabor, wavelet) and statistical descriptors (Haralick features based on gray-level co-occurrence matrix (GLCM), local binary patterns) to characterize optical texture and SAR scattering characteristics. And the fusion strategies are primarily founded on detail injection or guided filtering based on linear assumptions and prior rules [9], which have the advantage of demonstrating certain effectiveness in structural alignment but perform inadequately when addressing the complex nonlinear relationships between modalities [10]. Recent advancements in deep learning have notably surpassed traditional methods due to their end-to-end modeling capabilities and superior capacity to express high-dimensional features [11]. However, the application of a shared convolutional neural network (CNN) or transformer for both optical and SAR modalities neglects modality heterogeneity, resulting in feature confusion and a decline in segmentation accuracy [12]. To mitigate these limitations, deeper CNNs

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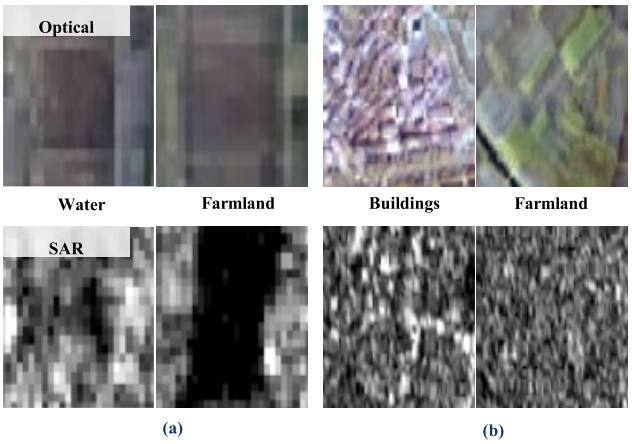


Fig. 1. Complementary representations in optical and SAR imagery. (a) SAR backscattering differences for water and farmland. (b) Optical texture differences between buildings and farmland.

and self-attention mechanisms have been employed to capture long-range dependencies. Nevertheless, these methods do not explicitly model non-Euclidean structural relationships; consequently, their generalization capability may be compromised. The discrete pixel domain of most existing fusion methods overlooks the continuous spatial characteristics of remote sensing scenes, thereby making fine details difficult to recover. Furthermore, spatial-domain-only methods neglect frequency-domain complementarity, which further restricts fusion accuracy [9]. The cross-domain perception strengthening can be performed by increasing the depth or width of the network or introducing the global attention mechanism in transformers, while the substantial rise in parameter count limits the generalization to unseen structures and gradually decreases the fusion accuracy [13], [14].

Graph convolutional networks (GCNs) explicitly model semantic associations and structural dependencies in non-Euclidean space through feature weighting in the Laplacian spectral domain [15]. This mechanism not only enables the analysis of the nonlinear coupling between geometric boundaries and scattering patterns in SAR images but also serves as a structural prior for cross-modal feature alignment, which enhances the discriminative power of optical and SAR semantic segmentation. Nevertheless, conventional GCNs are inherently restricted by their dependence on discrete graph construction and lack the capacity to preserve fine-grained spatial details. The mapping of implicit neural representation (INR) from discrete data to continuous space enables the precise capture of image details and overcomes the limitations of traditional discrete methods. It also offers a novel approach for high-fidelity image reconstruction and fine feature extraction [16]. However, the topological differences between optical and SAR modalities without geometric priors make it difficult for pure MLP-INR to align. In contrast, a graph-driven INR, which incorporates structural priors into the coordinate-to-semantic mapping, has shown strong potential for multimodal segmentation. To address these challenges, we revisit multimodal fusion from the perspective of structure-aware learning and continuous-domain representation. Inspired by the capacity of GCN to capture topological dependencies, we propose a dynamic adaptive GCN (DA-GCN) tailored to SAR's geometric properties. To further address spatial misalignment, we introduce graph-guided spatial implicit sampling (GSIS), embedding GCNs into the implicit coordinate representation to enhance spatial consistency. In addition, to compensate for missing high-frequency details in SAR's low-texture regions, we develop a graph-guided Fourier implicit sampling (GFIS) module to enable frequency-aware enhancement and global context modulation. We propose a spatial and frequency cooperative implicit fusion framework, termed MSFNet, for optical and SAR semantic segmentation. MSFNet unifies structure consistency, constrained alignment, and complementary detail enhancement under a single objective. It introduces a DA-GCN tailored to SAR geometric properties to enhance robust geometric structure and scattering pattern representations under speckle noise and scattering statistics variability. To mitigate spatial misalignment caused by geometric distortion and scale inconsistency, we propose GSIS, which injects structural priors in continuous coordinates to improve boundary and fine structure alignment. To compensate for missing high-frequency details in weak texture regions, we develop GFIS, which performs frequency domain implicit modulation to recover high-frequency content and reinforce global context consistency. These modules are integrated into a spatial-frequency implicit fusion framework, termed MSFNet (multimodal-interactivfe GCN-guided space–frequency implicit fusion network), to enable fine-grained cross-modal alignment and robust segmentation. To avoid ambiguity with prior uses of the acronym (e.g., multiple spatial fusion network [17]), we consistently refer to our method as MSFNet in the article. The main contributions are summarized as follows.

1) MSFNet is proposed as a spatial-frequency cooperative implicit fusion framework that jointly promotes fine-structure alignment and global-context enhancement to tackle cross-modal heterogeneity.
2) A dual-branch, nonshared CNN-GCN backbone is designed, where CNNs model optical texture/appearance, and DA-GCN enhances robust SAR geometric-structure and scattering-pattern representations.
3) GSIS is proposed to inject structural priors into continuous coordinates for improved alignment and boundary modeling, enhancing robustness to geometric distortion and scale inconsistency, particularly in deformation-prone regions.
4) GFIS is proposed to recover high-frequency details in weak-texture regions via frequency-domain implicit modulation while strengthening global contextual consistency.

The structure of this article is organized as follows. Section II introduces the relevant concepts of GCN and INR, followed by an overview of the development status of optical and SAR semantic segmentation. Section III presents the MSFNet architecture in detail and highlights the design of the DA-GCN and scattering-aware graph convolution (SAGC) modules. Section IV presents experimental results, compares MSFNet with representative baselines, and conducts ablation

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studies. Section V concludes the study and outlines potential directions for future research.

II. Related Work

A. Traditional Fusion for OSIs

Traditional methods for optical and SAR fusion can be broadly categorized into feature-level and data-level strategies. At the feature level, texture and scattering are modeled using handcrafted descriptors, such as Gabor filters for scale–direction patterns [18], GLCM statistics for structural regularity, and LBP for micro-texture encoding [19]. For SAR, preprocessing typically involves speckle suppression [20], followed by descriptors such as DWES [21] or polarimetric parameters including entropy-scattering angle [22] and Freeman–Durden decomposition [23]. While effective in capturing domain-specific features, these approaches rely heavily on predefined operators and strong priors, and thus are sensitive to misregistration, speckle, and background clutter, limiting their generalization in heterogeneous environments. Data-level fusion methods, on the other hand, are largely inspired by pixel-level strategies that combine the all-weather backscatter of SAR with the spectral richness of optical images [24]. These approaches are mainly grouped into component substitution (CS) and multiresolution analysis (MRA). CS methods, such as IHS-SAR, PCA-SAR, and Brovey-SAR [25], [26], [27], replace intensity or principal components of the optical image with registered SAR information, but often suffer from spectral distortion. Wavelet-SAR, Contourlet-SAR, and GSA-SAR [28], [29] exemplify MRA methods that retain low-frequency optical components and inject SAR high-frequency details using energy-or gradient-based weighting. This design yields better detail preservation but requires greater computational effort.

B. Deep Learning-Based Feature Fusion for OSI

Early multimodal semantic segmentation frameworks employed a dual-branch CNN structure [2], with multisource information fusion achieved through feature concatenation at the end of the encoder. Due to the locality of the convolutional receptive field, such designs struggle to capture long-range contextual dependencies between optical and SAR data [30]. ST-UNet introduces the Swin transformer [31] in the optical branch, aggregating contextual information through a local window attention mechanism. Window splitting disrupts the modeling of cross-window dependencies and significantly increases GPU memory consumption [31], and the integration of graph convolution into the encoder may address the geometric structure and scattering characteristics of SAR. Boundary-aware GCN (BGCN) constructs pixel-level boundary maps to enhance nontextural edges, while residual graph convolutional (RGC) employs residual graph convolution to preserve higher order topological relationships [32], [33]. Although GCNs are capable of propagating geometric information in non-Euclidean domains, the aforementioned methods still employ shared encoders to process both optical and SAR data simultaneously. Ignoring the differences in imaging mechanisms weakens the utilization of complementary information, thereby reducing object classification accuracy.

The prevailing deep learning-based fusion strategies can be broadly summarized as linear weighting, attention-based mechanisms, gating mechanisms, and explicit alignment [34]. Multimodal-cross attention network (MCANet) incorporates a cross-modal mutual attention module between dual CNN encoders and explicitly calculates channel correlations to strengthen fine-grained information, but still relies primarily on local convolutional kernels [34]. The dual-attention gating module [35] assigns learnable weights to features from both modalities, thereby mitigating noise and improving robustness, yet it lacks explicit modeling of geometric structures. TransFuse-RS employs geometric alignment blocks to enforce the spatial registration of OSIs, but overlooks frequency-domain information and suffers from high computational complexity [36]. Despite these advances, the majority of current approaches remain MLP-driven, with fusion mappings defined only in discrete spatial domains. Such designs fail to adequately account for modality-specific structural differences or continuous-space representations. Therefore, a spatial-frequency dual-domain continuous fusion framework with GCN-driven INR holds promise for simultaneously capturing optical texture and SAR scattering features, and offers a new direction toward higher accuracy and computational efficiency in multimodal remote sensing semantic segmentation.

C. Implicit Neural Representation

INR parameterizes continuous signals as neural functions, offering strong expressive power for modeling complex high-dimensional structures [37], [38]. Compared with explicit discrete representations, INR is more flexible in capturing intricate structures and latent relationships within data. Prior studies have shown that INR is effective in reconstructing high-resolution details for image super-resolution [14] and in fitting complex geometric shapes for 3-D reconstruction [39]. Recently, Hui et al. [40] extended INR to multimodal fusion and proposed a dual high-frequency framework leveraging cosine similarity, which significantly improved fusion performance. However, most of these approaches rely on MLP as the implicit mapping, and the independent Euclidean sampling strategy makes it difficult to capture structural relationships across modalities. In contrast, GNNs are inherently suitable for modeling non-Euclidean dependencies by establishing adjacencies and performing message passing. By replacing MLP with GNN as the implicit mapping, the proposed GNN-INR framework jointly models topological constraints and high-frequency details in continuous coordinate domains, thereby enhancing multimodal feature alignment and nonlinear mapping capacity.

D. GCN in Visual Tasks

GCN has been widely adopted in remote sensing semantic segmentation due to its ability to model structural dependencies in non-Euclidean spaces [15]. Existing approaches mainly focus on pixel-level graph construction, boundary enhancement, and attention-based fusion. For example, DSSN-GCN

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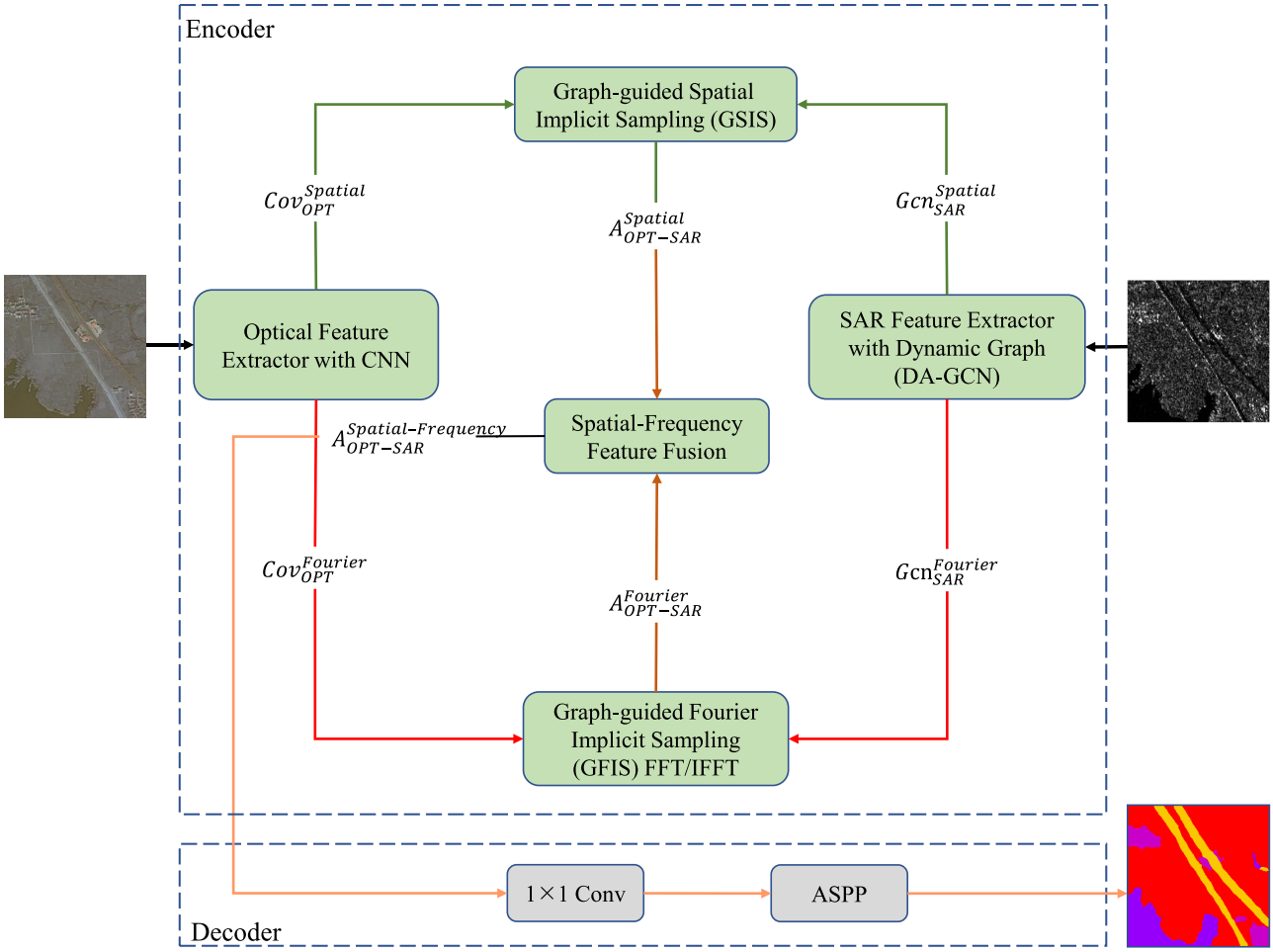


Fig. 2. Structure of MSFNet.

constructs spatial graphs to capture contextual correlations and enhance boundary discrimination [41]. S2-GCN further integrates contrastive learning into graph convolution to distinguish morphologically similar but semantically different land-cover classes [42]. Building on this trend, RGC stacks residual GCN layers on CNN features to jointly model local and global spatial relationships, whereas GAGUNet [43] and ViG [44] combine graph attention mechanisms to model long-range dependencies, achieving notable performance gains in both remote sensing and medical segmentation.

III. Methodology

A. Overall Architecture of MSFNet

MSFNet takes co-registered OSI pairs as input and extracts modality-specific multiscale features using a dual-branch encoder, as illustrated in Fig. 2. In the spatial domain, GSIS performs structure prior guided implicit sampling in continuous coordinates to align and reconstruct SAR features guided by the optical-branch topology. The features are further enhanced in the frequency domain by GFIS through implicit modulation to compensate for missing high-frequency details in weak texture regions and to strengthen global context consistency. The enhanced spatial and frequency representations are fused and decoded to produce the semantic prediction map.

The proposed MSFNet takes optical images (RGB or IRRG) and SAR images as input modalities. The optical branch adopts the pretrained ResNet-101 as the backbone, while the SAR branch employs the proposed DA-GCN to model backscatter features through an adaptive neighborhood approach. Each backbone preserves the first four layers as encoders, generating optical features $\mathbf{r}_i$ $(i = 1, 2, 3, 4)$ and SAR features $\mathbf{s}_i$ $(i = 1, 2, 3, 4)$, respectively. Taking a 256 × 256 input as an example, the resolution of the four-level feature maps decreases to 128×128, 64×64, 32×32, and 16×16, with corresponding channel dimensions of 64, 128, 256, and 512. Following the backbone, two implicit modules, namely GSIS and GFIS, are sequentially integrated to perform cross-modal alignment and information compensation in both the continuous spatial domain and the frequency domain. The overall architecture of MSFNet is illustrated in Fig. 3. GSIS and GFIS operate on the same features but use different domain priors. Their outputs are combined through a base-residual formulation. GSIS performs structure-guided implicit sampling in the continuous coordinate domain and outputs F_{spa}, which prioritizes geometric coherence and boundary stability under cross-modal structural mismatch. This structure-first sampling can attenuate high-frequency textures and global periodic components, especially in SAR regions dominated

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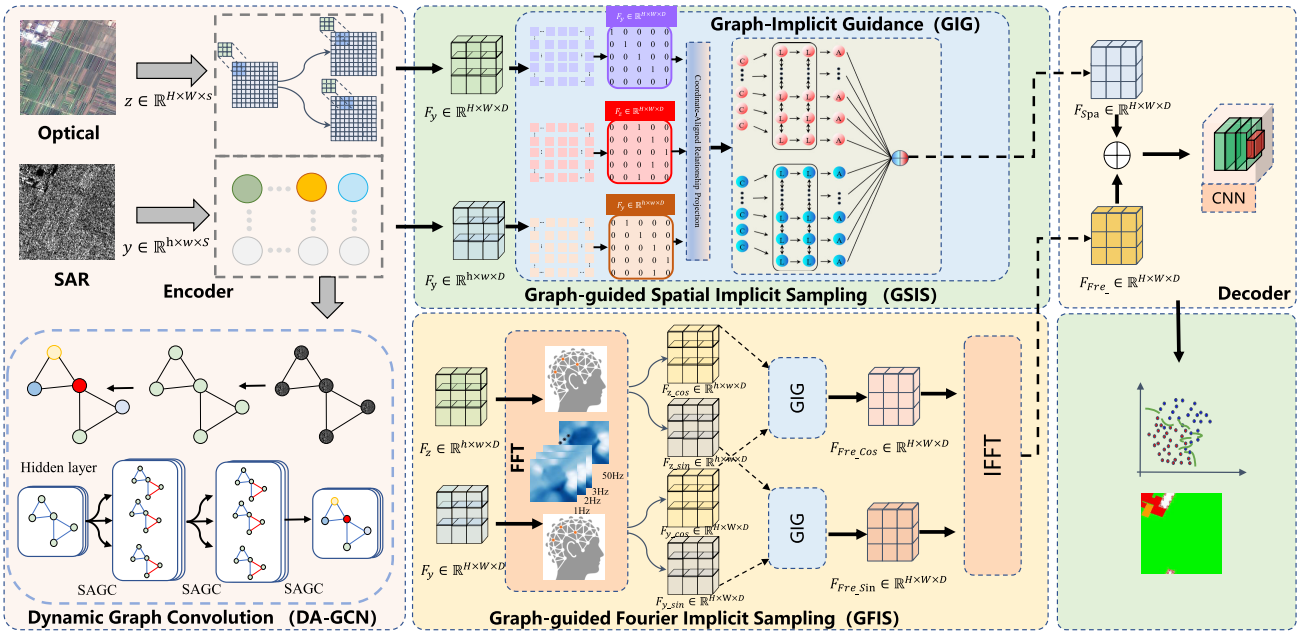


Fig. 3. Detailed architecture of MSFNet.

by speckle and low texture. GFIS therefore constructs a spectral correction F_{fre} by guiding the cosine/sine Fourier components under the same structure prior and restoring them via inverse Fourier transform, targeting the under-represented frequency components after spatial sampling. The additive fusion $F = F_{\text{spa}} + F_{\text{fre}}$ is thus interpreted as base–residual compensation. Under the constraints of spatial structure, F_{spa} is the structure-aligned spatial feature from GSIS that provides a structure-consistent base, F_{fre} is the spectral correction reconstructed by GFIS (after IFFT) that selectively compensates missing high-frequency and global-context components.

MSFNet employs a dual-branch encoder for feature extraction from OSIs, integrating the spatial domain GSIS and the frequency domain GFIS modules to enhance the synergistic modeling of cross-modal structures and semantics. Specifically, GSIS constructs graph adjacencies from the optical features in the continuous spatial domain to guide SAR features for structural alignment and reconstruction, thereby enhancing object contours and boundary perception. GFIS maps dual-path features into the Fourier domain, where graph-guided sampling in the sine-cosine subspace enriches frequency-domain semantics and models global periodic structures. Finally, the decoder sequentially integrates the fused spatial-frequency features, incorporating an atrous spatial pyramid pooling (ASPP) module in the first layer. By employing dilated convolutions with different dilation rates, the ASPP module captures multiscale contextual information, improving the ability to recover spatial details, strengthening the analysis of complex object boundaries, and ultimately increasing segmentation accuracy and robustness.

B. GCN Backbone for SAR (DA-GCN)

To improve the representation of complex scattering structures in SAR images, this study combines topological structures with backscatter statistical priors to develop a scattering-aware mechanism tailored for the graph construction and propagation stages. The mechanism enhances robustness against unstructured noise and multiscattering responses, providing structurally coherent graph features to facilitate spatial-frequency domain fusion.

In DA-GCN, an SAGC module is designed. First, the SAR image is divided into N nonoverlapping patches, and the mean μ_i and standard deviation σ_i of the backscatter for each patch are computed, which are embedded as node statistical features. Subsequently, a joint weighted metric function is constructed based on texture similarity, scattering differences, and spatial proximity, and a neighboring graph $G = (V, E)$ is built using dynamic KNN. Here, the node set V represents patches, and the edge set E represents the K-nearest neighbor connections. The edge weight between nodes is calculated by the combined distance d_{ij}

$$d_{ij} = \lambda_1\|x_i - x_j\|_2 + \lambda_2\|\mu_i - \mu_j\|_2 + \lambda_3\|\sigma_i - \sigma_j\|_2 + \lambda_4\|c_i - c_j\|_2 \quad (1)$$

where x_i, x_j represent image features, $\mu_i, \mu_j, \sigma_i, \sigma_j$ represent the statistical means and standard deviations of backscattering, c_i denotes the 2-D central coordinates of nodes, and $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are the weighting coefficients for respective distance components. $\lambda_1 \sim \lambda_4$ are the weighting coefficients for the distance components in the DA-GCN graph construction; we set $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 1$ in all experiments.

Dynamic KNN-based selection of K nearest neighbors according to d_{ij} constructs the graph $G = (V, E)$. Edge weights are defined as

$$w_{ij} = e^{-d_{ij}} \quad (2)$$

$$\alpha_{ij} = \frac{w_{ij}}{\sum_{k \in N(i)} w_{ik}} \quad (3)$$

where w_{ij} represents the original edge weight between nodes i and j, and α_{ij} represents the normalized edge weight. The

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scattering difference vector $\Delta s_{ij} = [\mu_i - \mu_j,\ \sigma_i - \sigma_j]$ modulates neighborhood aggregation to yield updated features

$$z_i = \sum_{j \in N(i)} \alpha_{ij} \cdot g\left(\left[x_i, x_j, \Delta s_{ij}\right]\right) \tag{4}$$

$$x_i' = x_i + W_h z_i \tag{5}$$

where $g(\cdot)$ denotes the neighborhood feature aggregation function, and the definitions of x_i and x_j follow those on the previous page; z_i represents the aggregated feature obtained after neighborhood aggregation, x_i' is the updated feature of node i, and W_h is the linear mapping matrix.

C. Graph-Implicit Guidance

Graph-implicit guidance (GIG) reconstructs SAR features in the continuous coordinate domain. In contrast to conventional point-wise implicit representations, GIG introduces graph-based message passing on an optical-guided structure prior to model interpolation weights, thereby explicitly incorporating geometric and topological constraints during the sampling stage.

Let the high-resolution optical features be denoted as $F_{\mathrm{OPT}} \in \mathbb{R}^{H \times W \times B_1}$, which serve as the reference. The low-resolution SAR features are denoted as $F_{\mathrm{SAR}} \in \mathbb{R}^{h \times w \times B_2}$, with $h \times w = H/4 \times W/4$, and are reconstructed in the continuous coordinate domain $\mathcal{C} = \{c_q\} \subset \mathbb{R}^2$.

The pixels of F_{OPT} are treated as nodes, and a k-neighborhood graph $G_{\mathrm{OPT}} = (V, E)$ is constructed under feature similarity and geometric proximity constraints, where the vertex set V corresponds to high-resolution pixels. For any query coordinate c_q, its bilinear neighborhood $\{x_a\}$ is located in the low-resolution SAR grid, and the k optical neighbors $\{x_i\}$ connected to node q in G_{OPT} are retrieved.

In GIG, the objective is to estimate the guided representation at each query coordinate c_q in a continuous domain while explicitly leveraging local structure in the high-resolution optical space. A point-wise MLP implements a mapping of the form $f_\theta(c_q)$ [or $f_\theta(c_q, F_{\mathrm{SAR}}(x_a))$], which treats queries independently and offers no explicit mechanism to enforce neighborhood-conditioned consistency. As a result, neighborhood topology and structural-consistency priors are not explicitly encoded during implicit sampling, limiting direct control over intraregion coherence and boundary stability. In contrast, we formulate coordinate-to-semantic estimation as structure-conditioned inference on a fixed k-neighbor set retrieved from the structure-guided graph G_{OPT}. This modeling assumption injects neighborhood consistency as a means of structural inductive bias and makes each query prediction explicitly constrained by local topology

$$z_q = g_\theta\left(c_q \mid \{x_i\}_{i=1}^{k}, G_{\mathrm{OPT}}, F_{\mathrm{OPT}}, F_{\mathrm{SAR}}\right) \tag{6}$$

where c_q denotes the query coordinate and z_q is the guided representation estimated at c_q; G_{OPT} is the structure-guided graph; $\{x_i\}_{i=1}^{k}$ denotes the fixed k neighbors retrieved on G_{OPT} (in all experiments, k is set to 8); $F_{\mathrm{OPT}}(\cdot)$ and $F_{\mathrm{SAR}}(\cdot)$ are the feature extractors for the optical and SAR branches, respectively, and x_a is the SAR anchor/reference location associated with the query. Under this conditional formulation, message passing aggregates neighborhood evidence on G_{OPT}, so the representation at c_q is jointly determined by its k-neighbor structure rather than by a single point input. This promotes intraregion semantic coherence under cross-modal gaps and local perturbations, while preserving structural discontinuities encoded by the graph connectivity. We construct $G_{\mathrm{OPT}} = (V_{\mathrm{opt}}, E_{\mathrm{opt}})$ over the optical feature map, where node i corresponds to pixel coordinate c_i with feature $f_i = F_{\mathrm{OPT}}(c_i)$. Edges are formed by coordinate-space KNN; each node is connected to its k nearest neighbors under 2-D geometric proximity.

TABLE I
SENSITIVITY ANALYSIS OF THE NEIGHBORHOOD SIZE k IN G_{OPT} ON THE WHU-OPT-SAR DATASET

k	OA(%)	Kappa(%)	mIoU(%)
4	81.43	73.94	46.70
8	82.11	74.62	54.28
12	81.50	73.97	48.11

A point-wise MLP treats each query independently and lacks an explicit mechanism for enforcing neighborhood/topology consistency. Although such a design can still learn effective implicit mappings, its robustness may be limited when SAR speckle and cross-modal appearance discrepancies weaken local structural coherence. To alleviate this issue, GCN message passing is introduced in the INR branch as a structure-aware inductive bias to promote neighborhood consistency and improve robustness. Subsequently, $F_{\mathrm{OPT}}(x_i)$, $F_{\mathrm{SAR}}(x_a)$, and the geometric offset $c_q - c_i$ are concatenated and fed into a single-layer GCN to predict the query vector

$$v_{q,i} = \mathrm{GCN}\left(\mathrm{cat}\left(F_{\mathrm{OPT}}(x_i),\ F_{\mathrm{SAR}}(x_a),\ c_q - c_i\right)\right) \in \mathbb{R}^{B_2+1} \tag{7}$$

where the first B_2 dimensions correspond to the spectral estimation, and the last dimension corresponds to the uncertainty. Here, x_a is the bilinear neighborhood index of c_q in the low-resolution SAR grid, x_i is the ith optical neighbor connected to node q in G_{OPT}, cat$(\cdot)$ denotes feature concatenation, and GCN$(\cdot)$ represents a single-layer graph convolution. Graph-convolution inputs comprise multisource features together with the relative geometry between a query and its neighbors. Optical features, SAR features, and relative coordinate encodings are concatenated to encode modality complementarity and query-neighbor spatial relations, yielding the node representation for graph aggregation

$$X_{q,i} = \mathrm{cat}\left(F_{\mathrm{OPT}}(x_i),\ F_{\mathrm{SAR}}(x_a),\ c_q - c_i\right) \tag{8}$$

where $F_{\mathrm{OPT}}(x_i)$ and $F_{\mathrm{SAR}}(x_a)$ are optical/SAR features at the corresponding locations, c_q and c_i denote the query and neighbor coordinates (or positional encodings), and cat$(\cdot)$ denotes channel-wise concatenation. In (6), GCN$(\cdot)$ aggregates these node features over the optical-guided graph G_{OPT}. We adopt the Kipf–Welling first-order spectral GCN, which enables one-hop propagation via self-loops and symmetric adjacency normalization. No attention mechanism or higher order spectral expansion is used, while local structural constraints are preserved during neighborhood aggregation. The propagation is given by

$$\mathrm{GCN}\left(X_{q,i}\right) = \sigma\left(\hat{A} X_{q,i} W\right) \tag{9}$$

$$\hat{A} = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}, \quad \tilde{A} = A + I \tag{10}$$

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TABLE II
QUANTITATIVE EVALUATION OF SEMANTIC SEGMENTATION RESULTS ON THE YYX-OPT-SAR DATASET

Method	Per-class Accuracy (%)						Evaluation Metric		
	Background	Bare land	Vegetation	Forest	Building	Roads	OA	Kappa	mIoU
MCANet	74.38	88.00	80.95	27.11	86.66	68.99	84.62	70.96	41.42
FTransUNet	81.65	85.82	74.55	22.82	85.76	68.84	82.56	66.84	39.98
SOFANet	69.70	88.27	72.78	27.69	80.72	73.65	82.62	68.04	39.18
JoiTriNet-d	64.23	82.05	78.68	34.93	81.34	46.33	79.91	60.36	33.28
DDHRNet	68.86	84.95	73.77	30.81	77.81	62.68	80.78	63.51	36.02
CFNet	72.61	84.28	77.20	30.11	81.35	70.44	81.66	64.69	38.50
PSCNN	74.25	85.22	75.05	49.58	84.63	71.18	82.27	66.14	39.91
DDP	73.44	83.85	73.22	25.00	82.11	56.41	80.24	62.21	35.76
CCEnd-Net	78.32	84.76	82.50	26.24	83.72	68.32	83.19	67.12	47.84
ASANet	83.70	86.76	79.17	32.07	84.93	75.14	84.40	70.04	40.79
MSFNet_OPT	77.70	85.98	80.06	35.46	70.10	60.43	82.46	66.73	37.97
MSFNet_SAR	0.00	63.70	0.00	0.00	0.00	0.00	63.59	0.26	7.96
MSFNet	75.85	88.73	80.14	38.86	87.63	76.21	85.52	72.83	43.44

TABLE III
QUANTITATIVE EVALUATION ON THE WHU-OPT-SAR DATASET

Method	Per-class Accuracy (%)							Evaluation Metric		
	farmland	city	village	water	forest	Roads	others	OA	Kappa	mIoU
MCANet	81.14	70.82	64.92	80.86	87.55	58.23	55.03	81.81	74.22	53.45
FTransUNet	79.94	74.67	67.78	81.77	88.45	57.82	50.47	81.92	74.35	47.73
SOFANet	81.01	63.71	65.86	81.80	88.26	56.92	54.66	81.56	73.95	46.57
JoiTriNet-d	78.84	60.52	58.95	78.58	83.85	41.93	43.34	77.93	68.71	40.85
DDHRNet	80.41	69.77	68.01	80.22	88.67	59.22	50.18	81.89	74.35	54.21
CFNet	80.07	60.07	64.98	81.53	85.16	55.78	46.98	79.67	71.16	42.19
PSCNN	81.26	64.70	69.02	81.96	81.93	54.89	55.77	79.83	71.18	50.39
DDP	80.24	65.58	68.41	81.30	87.19	53.97	51.16	81.17	73.30	45.80
CCEnd-Net	74.14	74.45	64.64	82.98	90.07	59.27	53.31	79.39	70.81	46.13
ASANet	80.32	60.82	66.10	78.62	87.81	59.17	49.50	80.22	72.22	48.48
MSFNet_OPT	77.49	70.44	63.21	81.48	87.91	59.25	51.10	80.23	72.00	44.24
MSFNet_SAR	64.14	33.76	30.85	73.24	86.65	33.98	0.00	66.08	53.69	27.48
MSFNet	81.61	69.66	67.48	82.85	87.01	60.24	51.80	82.11	74.62	54.28

TABLE IV
QUANTITATIVE EVALUATION OF SEMANTIC SEGMENTATION RESULTS ON THE DFC23 DATASET

Method	Per-class Accuracy (%)													Evaluation Metric		
	BG	FR	GR	GR2	RR	MER	HRv1	HRv2	MR	PR	AR	Do	OR	OA	Kappa	mIoU
MCANet	94.34	76.32	62.11	70.82	74.89	27.58	70.47	53.95	61.36	24.78	43.20	53.24	60.58	89.53	70.88	32.64
FTransUNet	93.48	73.51	54.52	50.16	72.00	18.88	0.00	50.06	0.00	32.00	0.00	0.00	0.00	88.38	66.89	19.26
SOFANet	93.76	72.45	55.63	65.79	73.06	0.00	95.00	55.39	74.00	0.00	48.32	78.87	64.77	88.56	67.78	26.11
JoiTriNet-d	89.50	68.89	49.59	52.45	68.07	0.00	92.35	41.54	40.18	0.11	43.43	0.00	41.27	85.15	55.42	17.30
DDHRNet	94.32	75.09	58.61	58.93	71.25	19.41	72.85	43.69	85.44	21.09	38.94	59.08	52.79	89.00	69.46	61.06
CFNet	89.07	65.44	49.32	50.43	41.53	11.57	0.00	42.83	0.00	6.00	0.00	0.00	1.09	84.68	52.87	14.27
PSCNN	91.16	66.24	52.44	48.35	47.51	0.00	0.00	42.65	0.00	0.00	0.00	0.00	0.00	86.06	58.88	15.15
DDP	93.50	69.32	51.02	57.64	53.55	0.00	0.00	61.79	0.00	0.00	0.00	0.00	26.81	87.46	64.86	17.98
CCEnd-Net	89.72	75.42	38.96	33.14	27.28	0.00	70.81	46.77	0.00	0.00	0.00	0.00	31.73	83.73	51.34	16.14
ASANet	92.62	67.61	49.29	64.47	64.69	0.00	0.00	52.86	0.00	0.00	0.00	0.00	23.90	86.85	62.34	17.83
MSFNet_OPT	84.22	57.59	57.71	46.63	0.00	88.93	49.84	0.00	0.00	1.41	0.00	0.00	32.60	87.25	33.89	13.90
MSFNet_SAR	37.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	77.83	0.00	0.00
MSFNet	94.92	76.97	62.52	67.66	57.41	31.12	65.97	58.71	62.99	28.81	50.35	80.51	56.27	89.87	72.15	33.89

where W is a learnable linear map and $\sigma(\cdot)$ denotes the activation; A is the adjacency of G_{OPT}, I is the identity (self-loops), and $\tilde{D}$ is the degree matrix of $\tilde{A}$ with $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Here, the graph convolution in (6) serves one-hop aggregation for interpolation queries rather than deep graph reasoning. Hence, a single-layer GCN is used to enforce local geometric and topological consistency. Deeper GCN stacks are not explored, since they add complexity and may weaken locality, which

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TABLE V
QUANTITATIVE EVALUATION OF SEMANTIC SEGMENTATION RESULTS ON THE DFC2025 DATASET

Methods	Per-class Accuracy (%)								Evaluation Metric		
	Bareland	Rangeland	Developed space	Road	Tree	Water	Agriculture land	Building	OA	Kappa	mIoU
MCANet	51.71	83.82	74.32	77.99	89.40	95.34	80.08	83.69	82.39	78.80	69.47
FTransUNet	56.40	83.71	74.23	78.63	89.21	95.32	84.97	87.00	83.22	79.81	69.50
SOFNet	64.12	84.18	74.53	73.60	85.82	96.07	85.40	87.02	82.72	79.18	69.65
JoiTriNet-d	60.24	82.92	70.92	72.04	81.17	93.77	82.12	82.97	79.85	75.75	66.52
DDHRNet	55.77	85.22	76.77	72.20	86.70	96.11	82.30	86.92	83.20	79.81	70.36
CFNet	50.36	81.07	71.58	68.43	84.25	94.85	81.04	79.08	79.28	75.06	65.57
PSCNN	44.34	71.63	66.77	73.91	88.32	90.87	82.85	83.62	76.13	70.88	59.42
DDP	65.94	83.51	73.73	72.66	87.23	94.96	83.61	84.39	82.10	78.43	67.87
CCEnd-Net	43.08	79.10	68.67	66.52	82.71	94.69	85.86	88.78	78.45	73.93	64.20
ASANet	51.13	82.80	75.70	76.23	87.28	96.15	84.50	85.58	82.67	79.13	70.04
MSFNet_OPT	61.75	82.54	74.45	74.91	88.72	96.79	81.32	84.97	82.42	78.79	63.47
MSFNet_SAR	0.00	40.99	39.30	60.41	43.61	36.25	26.55	46.22	39.64	26.91	39.64
MSFNet	61.45	84.18	75.53	76.82	88.88	97.01	80.69	87.99	83.51	80.12	70.61

TABLE VI
PROGRESSIVE ABLATION RESULTS OF MSFNET ON THE YYX-OPT-SAR DATASET

Methods	DA-GCN	GSIS	GFIS	OA	mIoU	Kappa	Model Size (MB)	Params (M)	FLOPs (G)
CNN only	×	×	×	81.01	41.47	66.83	449.34	117.58	27.75
CNN + DA-GCN	✓	×	×	78.95	40.53	63.90	387.78	101.53	22.58
CNN + DA-GCN + GSIS	✓	✓	×	78.69	39.99	63.98	396.54	103.83	23.17
CNN + DA-GCN + GFIS	✓	×	✓	77.56	39.01	62.20	391.30	102.46	22.82
MSFNet	✓	✓	✓	**85.52**	**43.44**	**72.83**	**332.46**	**87.08**	**17.76**

TABLE VII
ADDITIONAL ABLATION AND VARIANT ANALYSIS OF MSFNET ON THE YYX-OPT-SAR DATASET

Methods	DA-GCN	GCN	GSIS	GFIS	OA	mIoU	Kappa	Modelsize (MB)	Params (M)	FLOPs (G)
w/o DA	×	✓	✓	✓	80.70	35.92	63.86	227.11	59.42	16.54
w/o GF	✓	✓	✓	×	82.15	38.37	66.77	323.90	84.84	17.71
w/o GC	✓	×	✓	✓	80.09	42.26	65.65	404.06	105.80	24.00
w/o GS	✓	✓	×	✓	82.00	38.73	66.22	318.67	83.47	17.71
w/o GS & GF	✓	✓	×	×	83.41	39.88	68.97	327.44	82.54	17.71
MSFNet	✓	✓	✓	✓	85.52	43.44	72.83	332.46	87.08	17.76

is undesirable for interpolation. Specifically, $v_{q,i}[1:B_2]$ is the predicted B_2-dimensional spectral vector, and $v_{q,i}[B_2]$ is the corresponding uncertainty scalar.

The k query vectors are concatenated into $V_q \in \mathbb{R}^{k(B_2+1)}$, and the uncertainties are normalized with a softmax function to obtain the weight $\alpha_{q,i}$. The final guided sampling result is given by

$$z_q = \sum_{i=1}^{k} \alpha_{q,i}\, v_{q,i}\,[1:B_2] \tag{11}$$

$$\alpha_{q,i} = \text{softmax}\left(v_{q,i}\,[B_2]\right) \tag{12}$$

where $z_q \in \mathbb{R}^{B_2}$ is the guided sampling result at coordinate c_q, and $\alpha_{q,i}$ is the adaptive weight based on uncertainty normalization, satisfying $\sum_{i=1}^{k} \alpha_{q,i} = 1$. Here, k denotes the number of nearest neighbors used to construct the KNN graph. In our implementation, we fix $k = 8$ to ensure stable performance, as supported by the sensitivity analysis in Appendix, Table I.

D. Graph-Guided Spatial Implicit Sampling

GSIS leverages the structural features extracted by the optical branch as spatial priors to guide the implicit alignment and fusion of SAR features within the continuous coordinate domain. A high-resolution coordinate grid is first constructed based on the pixel centers $\mathcal{C}$ of the optical feature map. Subsequently, a structure-guided graph G_{OPT} is generated from feature similarity and geometric proximity, explicitly encoding the spatial topology of optical edges and textures as constraints to guide the adaptive sampling of F_{SAR} in GIG.

For each query coordinate, weights are dynamically assigned based on the graph neighborhood to preserve fine-grained high-frequency details while mitigating feature aliasing and boundary blurring that typically arise in traditional interpolation. This improves the structural consistency across modalities. The formulation is as follows:

$$F_{\text{fused}} = \text{GIG}\left(F_{\text{SAR}}, F_{\text{OPT}}, G_{\text{OPT}}, \mathcal{C}\right) \tag{13}$$

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TABLE VIII
COMPARISON OF COMPUTATIONAL COMPLEXITY AND SEGMENTATION PERFORMANCE

Methods	Modelsize (MB)	Params (M)	FLOPs (G)	Latency (ms)	FPS (patches/s)	Peak Mem (MB)	mIoU (%)
MCANet	358.15	93.78	20.81	27.30	36.63	541.90	41.42
DDP	359.27	93.07	85.01	29.627	33.75	554.30	35.76
PSCNN	25.62	6.71	44.23	5.76	173.62	222.60	39.91
CFNet	2.15	0.55	2.38	6.84	146.21	148.50	38.50
DDHRNet	126.48	33.10	14.79	29.97	33.37	538.00	36.02
SOFANet	241.14	63.20	70.12	7.79	128.38	862.60	39.18
JoiTriNet-d	335.88	87.84	20.63	29.25	34.19	1442.50	33.28
FTransUNet	775.94	203.40	59.75	44.75	22.35	1703.10	39.98
CCEnd-Net	7.85	2.05	67.16	14.86	67.29	517.90	47.84
ASANet	120.00	31.44	41.82	19.74	50.65	707.40	40.79
MSFNet	332.46	87.08	17.76	192.43	5.20	1379.70	43.44

TABLE IX
SENSITIVITY ANALYSIS UNDER CONTROLLED RESIDUAL CROSS-MODAL SHIFTS

RegLevel	Metrics	CCEnd-Net	ASANet	MCANet	PSCNN	DDP	DDHRNet	FTransUNet	SOFANet	JoiTriNet-d	CFNet	MSFNet
Aligned	OA	83.19	84.10	84.62	82.27	80.24	80.78	82.56	82.62	79.91	81.66	85.52
	mIoU	38.28	39.96	41.42	39.91	35.76	36.02	39.98	39.18	33.28	38.50	43.44
	mBF	20.03	22.15	23.63	19.89	13.77	13.80	23.59	21.22	9.05	18.73	25.00
Low	OA	83.18	84.11	84.52	81.14	80.22	80.54	82.96	82.63	79.58	81.46	85.39
	mIoU	38.27	39.99	41.88	38.79	35.81	35.69	40.04	39.11	33.95	38.60	43.49
	mBF	20.02	22.35	23.54	20.22	13.62	14.10	23.47	21.06	8.67	18.32	24.71
Medium	OA	83.15	84.09	84.49	81.13	80.24	80.33	82.85	82.60	79.71	81.33	85.33
	mIoU	38.24	39.92	41.78	38.79	35.84	35.52	39.71	39.13	34.12	64.98	43.36
	mBF	20.13	22.31	23.21	20.09	13.77	13.90	23.40	20.71	8.58	18.55	24.72
High	OA	83.16	84.09	84.52	81.09	80.23	80.41	82.47	82.54	79.51	81.29	85.28
	mIoU	38.23	39.99	41.80	38.68	35.68	35.63	39.43	39.03	33.78	38.42	43.24
	mBF	20.00	22.43	23.08	19.85	13.73	14.55	22.98	20.59	8.44	18.66	24.57

Notes: Aligned/Low/Medium/High correspond to SAR shifts $k = \{0, 2, 5, 8\}$ px. The SAR input is translated by $(\Delta x, \Delta y) = (k, 0)$ while optical images and labels are fixed; reflect padding is applied only to preserve the original input size. This setting is intended as a controlled perturbation test for residual cross-modal offsets rather than a full simulation of real registration errors.

where F_{fused} denotes the spatially guided features generated by GIG, F_{SAR} and F_{OPT} are the encoded SAR and optical features, respectively, G_{OPT} represents the structure-guided graph derived from the optical image, and $\mathcal{C}$ is the continuous coordinate set corresponding to the optical image.

E. Fourier-Based Implicit Sampling (GFIS)

GFIS utilizes a frequency-domain guidance mechanism to enhance the spectral representation of the fused image. The encoded SAR and optical features undergo 2-D Fourier transforms to obtain their complex-valued frequency-domain representations. The real and imaginary components are disentangled, with the real part corresponding to the cosine channel and the imaginary part corresponding to the sine channel. GFIS encodes the complex spectrum as two real-valued channels in the frequency domain, corresponding to the cosine and sine components. This provides an information-preserving and invertible $\mathbb{R}^2$ representation of complex coefficients; therefore, channel-wise processing introduces no additional assumptions and retains phase-encoded structural information

$$\hat{F}(\omega) = \hat{F}^{\cos}(\omega) + j\,\hat{F}^{\sin}(\omega) \tag{14}$$

where ω is the 2-D frequency coordinate, $\hat{F}(\omega) \in \mathbb{C}$ is the complex coefficient, $\hat{F}^{\cos}(\omega)$ and $\hat{F}^{\sin}(\omega)$ denote the real and imaginary parts, and j denotes the imaginary unit. Phase is uniquely determined by the cosine-sine pair, not by either branch alone. GFIS jointly reconstructs both components on the same frequency grid under identical guided sampling, recombines them into a complex spectrum, and applies IFFT for spatial-domain optimization. Phase inconsistency manifests as higher spatial reconstruction error and structural artifacts, and is penalized by spatial-domain supervision

$$\angle\hat{F}(\omega) = \operatorname{atan2}\left(\hat{F}^{\sin}(\omega),\, \hat{F}^{\cos}(\omega)\right) \tag{15}$$

where $\angle\hat{F}(\omega)$ denotes the phase of $\hat{F}(\omega)$, and $\operatorname{atan2}(\cdot,\cdot)$ denotes the four-quadrant arctangent.

It should be noted that GFIS is not formulated as an explicit SAR despeckling module, nor does it operate as a learned low-pass filter. In coherent SAR imaging, speckle and phase instability can introduce locally irregular spectral responses, especially in the high-frequency band. However, high-frequency components in SAR are not purely noise-dominated, since physically meaningful structures such as object bound-

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aries, thin linear targets, and abrupt geometric transitions are also encoded in this band. For this reason, uniformly suppressing high-frequency energy would inevitably weaken structurally informative responses together with speckle-related perturbations.

GFIS performs structure-constrained reconstruction of the complex spectrum through its cosine-sine representation. The optical branch provides a more stable structural prior, which is used to condition the reconstruction of SAR frequency responses. Under this guidance, frequency components that are consistent with cross-modal structural support are preserved or complemented, whereas locally unstable responses lacking such support are suppressed during reconstruction. Therefore, GFIS does not smooth the spectrum indiscriminately; rather, it regularizes SAR high-frequency responses according to cross-modal structural consistency.

In this sense, GFIS is more appropriately understood as a frequency-domain complementary compensation and consistency-regularization mechanism under structural guidance, rather than a generic frequency-smoothing operator.

Next, GIG operations are independently performed on the cosine and sine channels to produce cosine-guided and sine-guided features. These guided features are then integrated and restored into frequency-domain guided features through an inverse Fourier transform. By independently guiding the cosine and sine components, GFIS effectively compensates for high-frequency structural information loss in the SAR modality and achieves structural complementarity and energy consistency in the spectral domain when combined with the high-resolution optical features. The mathematical formulation of GFIS is given as

$$F_{\mathrm{SAR}}^{\cos}, F_{\mathrm{SAR}}^{\sin} = \mathrm{FFT}\left(F_{\mathrm{SAR}}\right) \tag{16}$$

$$F_{\mathrm{OPT}}^{\cos}, F_{\mathrm{OPT}}^{\sin} = \mathrm{FFT}\left(F_{\mathrm{OPT}}\right) \tag{17}$$

$$F_{\mathrm{fre}}^{\cos} = \mathrm{GIG}\left(F_{\mathrm{SAR}}^{\cos}, F_{\mathrm{OPT}}^{\cos}, \mathrm{Coord}\right) \tag{18}$$

$$F_{\mathrm{fre}}^{\sin} = \mathrm{GIG}\left(F_{\mathrm{SAR}}^{\sin}, F_{\mathrm{OPT}}^{\sin}, \mathrm{Coord}\right) \tag{19}$$

$$F_{\mathrm{fre}} = \mathrm{IFFT}\left(F_{\mathrm{fre}}^{\cos}, F_{\mathrm{fre}}^{\sin}\right) \tag{20}$$

where $F_{\mathrm{SAR}}^{\cos}, F_{\mathrm{SAR}}^{\sin}$ denote the cosine and sine components of the encoded SAR features, $F_{\mathrm{OPT}}^{\cos}, F_{\mathrm{OPT}}^{\sin}$ denote the cosine and sine components of the encoded optical features, $F_{\mathrm{fre}}^{\cos}$ and $F_{\mathrm{fre}}^{\sin}$ are the guided features in the frequency domain, and F_{fre} is the final fused frequency-domain representation. Coord denotes the set of coordinates for frequency-domain sampling.

F. Decoder Unit

The decoder integrates both spatial and frequency-domain guided features to restore pixel-level semantic labels under multiscale contextual constraints. An additive fusion approach is adopted to combine F_{spa} and F_{fre}, ensuring both structural alignment and spectral compensation. To further enhance contextual modeling, an ASPP module is incorporated, which captures multiscale contextual information using dilated convolutions.

The fused features are convolved along the channel dimension and upsampled to the original spatial resolution. Pixel-level semantic predictions are then generated via a Softmax layer. The overall decoding process can be expressed as

$$\hat{Y} = \mathrm{Softmax}\left(\mathrm{Upsample}\left(\mathrm{Conv}\left(\mathrm{ASPP}\left(F_{\mathrm{spa}} + F_{\mathrm{fre}}\right)\right)\right)\right) \tag{21}$$

where F_{spa} and F_{fre} are the dual-domain inputs, $\hat{Y}$ is the final semantic segmentation map, and C is the number of classes.

DA-GCN builds a dynamic KNN graph on patch-level nodes extracted from downsampled intermediate features. With node number N, fixed neighborhood size k, and feature dimension d, one message-passing layer has complexity approximately $O(NKd)$, and the extra memory is dominated by neighbor indices and edge weights with order $O(NK)$. Here, N is determined by the patch grid, and the computation is performed on intermediate-resolution features. GFIS applies 2-D FFT/IFFT to an intermediate D channels feature map of size $H' \times W'$, with complexity $O(DH'W' \log(H'W'))$. The cosine/sine branches correspond to two GIG-guided sampling paths and introduce mainly a constant-factor overhead in computation and intermediate storage. In practice, both DA-GCN and GFIS operate on intermediate-scale features, and we use a fixed k and tensor reuse to limit the extra cost.

G. Loss Function

The cross-entropy loss function is used as the primary loss metric to quantify the difference between the model's predicted probability distribution and the true label distribution, and to effectively capture the prediction confidence in multiclass classification. Compared with regression-based losses such as mean squared error (mse), cross-entropy excels in classification boundary clarity, gradient stability, and adaptability to class imbalance. This renders it particularly effective for remote sensing scenes with complex textures and interwoven class boundaries in semantic segmentation.

Algorithm 1 Proposed MSFNet

Require: SAR $\mathbf{SAR} \in \mathbb{R}^{H\times W\times s}$, Optical $\mathbf{OPT} \in \mathbb{R}^{h\times w\times S}$
Ensure: Semantic prediction map $\hat{\mathbf{Y}} \in \mathbb{R}^{H\times W\times C}$

for each training step **do**
  Step 1: Extract modality-specific features
  $\mathbf{F}_{\mathrm{SAR}} \leftarrow \mathrm{Encoder}(\mathbf{SAR})$
  $\mathbf{F}_{\mathrm{OPT}} \leftarrow \mathrm{Encoder}(\mathbf{OPT})$
  Step 2: GSIS
  $\mathbf{F}_{\mathrm{Spa}} \leftarrow \mathrm{GIG}(\mathbf{F}_{\mathrm{SAR}}, \mathbf{F}_{\mathrm{OPT}}, \mathcal{C})$
  Step 3: GFIS
  $(\mathbf{F}_{\mathrm{SAR}}^{\cos}, \mathbf{F}_{\mathrm{SAR}}^{\sin}) \leftarrow \mathrm{FFT}(\mathbf{F}_{\mathrm{SAR}})$
  $(\mathbf{F}_{\mathrm{OPT}}^{\cos}, \mathbf{F}_{\mathrm{OPT}}^{\sin}) \leftarrow \mathrm{FFT}(\mathbf{F}_{\mathrm{OPT}})$
  $\mathbf{F}_{\mathrm{Fre}}^{\cos} \leftarrow \mathrm{GIG}(\mathbf{F}_{\mathrm{SAR}}^{\cos}, \mathbf{F}_{\mathrm{OPT}}^{\cos}, \mathcal{C})$
  $\mathbf{F}_{\mathrm{Fre}}^{\sin} \leftarrow \mathrm{GIG}(\mathbf{F}_{\mathrm{SAR}}^{\sin}, \mathbf{F}_{\mathrm{OPT}}^{\sin}, \mathcal{C})$
  Step 4: Frequency Feature Fusion
  $\mathbf{F}_{\mathrm{Fre}} \leftarrow \mathrm{IFFT}(\mathbf{F}_{\mathrm{Fre}}^{\cos}, \mathbf{F}_{\mathrm{Fre}}^{\sin})$
  Step 5: Spatial–Frequency Feature Integration
  $\mathbf{F}_{\mathrm{Combined}} \leftarrow \mathbf{F}_{\mathrm{Fre}} + \mathbf{F}_{\mathrm{Spa}}$
  Step 6: Decoding
  $\hat{\mathbf{z}} \leftarrow \mathrm{Decoder}(\mathbf{F}_{\mathrm{Combined}})$
end for
return $\hat{\mathbf{Y}}$

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Pixel-wise cross-entropy is used as the optimization objective in the main experiments. Boundary and fine-structure preservation primarily stems from the structural design, and cross-entropy provides stable pixel-wise supervision that complements these constraints. Dice/IoU and boundary losses often require extra hyperparameters (e.g., weights and boundary bandwidth) and can be more sensitive to class imbalance and label noise; therefore, these auxiliary losses are omitted in the main experiments to ensure fair and reproducible comparisons. Accordingly, cross-entropy is used for optimization and is defined as

$$L_{\mathrm{CE}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\,\log\left(p_{i,c}\right) \tag{22}$$

where "CE" stands for cross entropy and L_{CE} denotes the standard cross-entropy loss; N represents the total number of pixels, C denotes the number of classes, $y_{i,c}$ is the ground-truth (GT) indicator of pixel i for class c, and $p_{i,c}$ is the predicted probability for class c at pixel i.

IV. Experiments

A. Datasets

This study selects four datasets, namely WHU-OPT-SAR, DFC23, YYX-OPT-SAR, and DFC2025, for experimentation to validate the superiority of MSFNet.These datasets cover a range of imaging sensors, spatial resolutions, and scene complexities, enabling the evaluation of the model's robustness and generalization ability under multimodal conditions.

The current public optical-SAR benchmarks do not provide standardized annotations for radar viewing geometry, and their coverage of strongly rugged terrain is also limited. Nevertheless, this does not eliminate the relevance of SAR geometry-related effects. In particular, line-of-sight misalignment, layover, and shadowing may still influence cross-modal consistency and segmentation performance, especially in geometrically complex scenes. Therefore, the current benchmark setting should be regarded as a standard comparative evaluation rather than a dedicated geometry-aware assessment, and the influence of SAR viewing geometry remains an important limitation of the present study. Future work will consider dedicated experiments on more geometrically complex SAR scenes, as well as the incorporation of digital elevation models (DEMs) into the graph construction process to better characterize terrain-dependent geometric distortions.

The public benchmark datasets mainly target commonly used semantic segmentation settings and may not comprehensively cover strongly rugged terrains. Moreover, they do not provide standardized per-pixel SAR viewing-geometry annotations (e.g., look angle or radar line-of-sight direction). Since SAR viewing geometry can affect the appearance of orientation-sensitive targets (e.g., roads and building edges), we follow the benchmark protocol for comparative evaluation and treat geometry-aware analysis on more topographically complex terrains as an important direction for future work. All optical and SAR pairs used in this study are taken as co-registered inputs, as provided in the dataset releases. The detailed descriptions of the datasets are provided below:

1) WHU-OPT-SAR Dataset: The WHU-OPT-SAR dataset [34] consists of 100 pairs of OSIs, each covering the same region, with both modalities having a spatial resolution of 5 m. The images are of dimensions 5556 × 3704 pixels and span approximately 50 000 km^2 in Hubei, China (30°N–33°N, 108°E–117°E). The dataset includes seven primary land cover classes: farmland, urban areas (city), villages (village), water bodies, forests, roads, and others. We follow the official land-use annotation scheme provided by the dataset without any relabeling. In particular, "city" and "village" correspond to different land-use categories, i.e., urban built-up areas and rural/township settlements, respectively. As the dataset is designed for land-use classification that emphasizes social attributes of land utilization, these two classes may both contain buildings but represent distinct land-use semantics. The 100 image pairs are randomly split into training, validation, and test sets in a ratio of 6:2:2. The training set contains 60 pairs of optical images, SAR images, and corresponding labels, while the validation and test sets contain 20 pairs each. Before network training, the original 5556 × 3704 images are cropped into 256 × 256 pixel patches. After cropping, the training set consists of 17 640 image pairs, while both the validation set and the test set contain 5880 pairs each.

2) DFC23: The DFC23 [45] is a multimodal rooftop classification dataset released for the 2023 IEEE GRSS Data Fusion Contest, consisting of paired SAR and optical imagery. It defines 12 semantic classes according to rooftop geometry (e.g., gabled, arched, and pyramid roofs) and is formulated as a pixel-level semantic segmentation task. The dataset is split into training/validation/test subsets with a 6:2:2 ratio, providing high-quality annotations and diverse scene variations. For clarity, the class abbreviations used in Fig. 4 and Table II follow the official DFC23 label definitions, and their full names are given in the caption of Fig. 4.

3) YYX-OPT-SAR: The YYX-OPT-SAR dataset [46] provides 150 co-registered optical-SAR image pairs, each with spatial dimensions of 512 × 512 pixels, covering urban, suburban, and mountainous zones. The SAR imagery was acquired by an uncrewed aerial vehicle (UAV) within the geographic range 110.135°E–110.160°E and 34.593°N–34.626°N, while the optical counterparts were obtained from Google Earth at a ground sampling distance (GSD) of 0.5 m. The dataset is split into training, validation, and test sets with a ratio of 9:3:3, containing 90, 30, and 30 image pairs, respectively. Before network training, all images are cropped to 256 × 256 pixels, resulting in 360 training pairs and 120 pairs each for validation and testing.

4) DFC2025 Dataset: DFC2025 refers to Track 1 of the 2025 IEEE GRSS Data Fusion Contest on all-weather land-cover mapping [49]. It provides approximately 4300 paired aerial RGB and SAR image tiles with 8-class land-cover pseudo-labels (bareland, rangeland, developed space, road, tree, water, agriculture land, and building), covering 35 regions in Japan, USA, and France, with a GSD in the range of 0.15–0.5 m. The pseudo-labels are generated using pretrained OpenEarthMap models based on optical imagery. It should be noted that the "GT" of DFC2025 is therefore a model-derived proxy rather than manual annotation, which may

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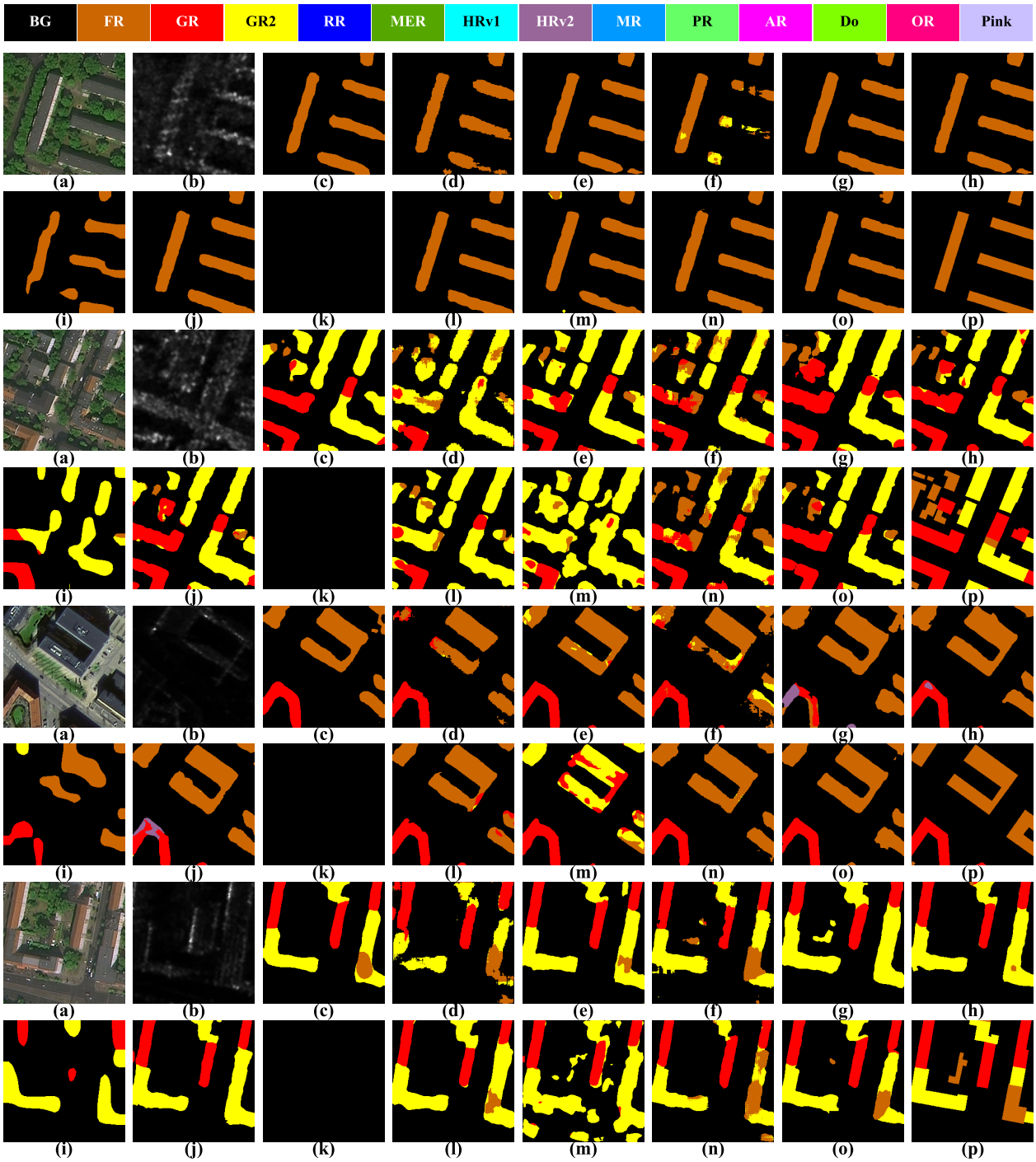


Fig. 4. Qualitative semantic segmentation results on the DFC23 dataset. The subfigures are arranged in the following order: (a) optical image, (b) SAR image, (c) DDP, (d) CFNet, (e) FTransUNet, (f) PSCNN, (g) MCANet, (h) SOFANet, (i) JoiTriNet, (j) DDHRNet, (k) SAR-only, (l) Opt-only, (m) CCEnd, (n) ASANet, (o) MSFNet, and (p) GT. The figure shows that the proposed method yields clearer roof-category boundaries and more accurate category discrimination in complex rooftop scenes. Class abbreviations: BG = Background; FR = Flat Roof; GR = Gabled Roof; GR2 = Gabled Roof (variant); RR = Round Roof; MER = Metal Roof; HRv1/HRv2 = Hipped Roof (type 1/type 2); MR = Mansard Roof; PR = Pyramid Roof; AR = Arched Roof; Do = Dome; OR = Other Roof; and Pink = Other Roof (following the official DFC23 definitions).

introduce systematic biases into the reported accuracy. While the official benchmark emphasizes SAR-based evaluation, our study targets optical-SAR fusion segmentation; thus, we use dual-modality inputs for both training and testing. We split the dataset into training, validation, and test sets with a ratio of 6:2:2, and crop all images into 256×256 patches, following the same preprocessing and evaluation protocol as the other datasets.

B. Evaluation Metrics and Implementation Details

1) Training Setup: All experiments were performed on a machine equipped with an NVIDIA 3080 GPU and based on

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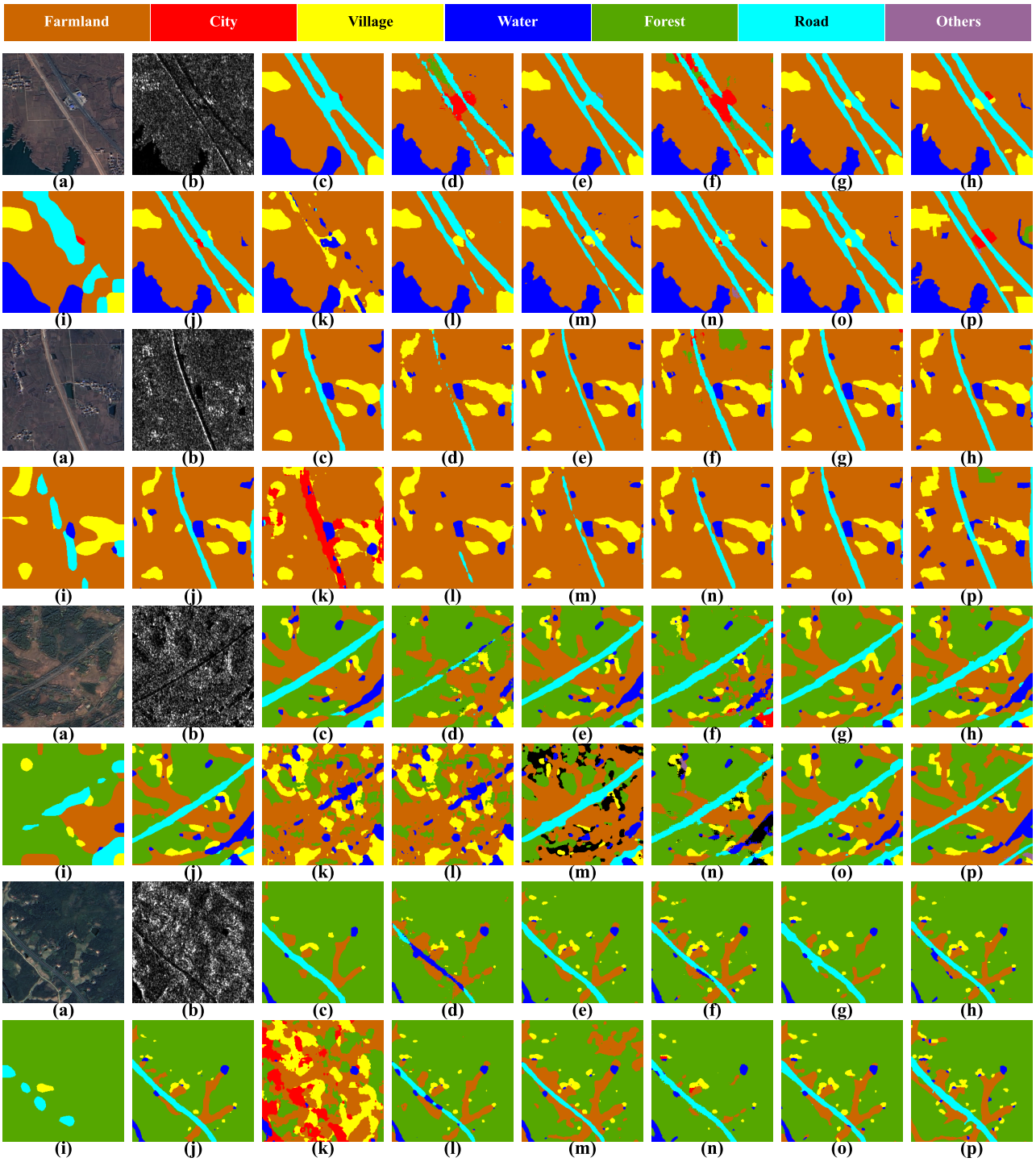


Fig. 5. Semantic segmentation results in the WHU-OPT-SAR dataset. (a) Optical. (b) SAR. (c) DDP. (d) CFNet. (e) FTransUNet. (f) PSCNN. (g) MCANet. (h) SOFANet. (i) JoiTriNet. (j) DDHRNet. (k) SAR-only. (l) Opt-only. (m) CCEnd. (n) ASANet. (o) MSFNet. (p) GT.

the PyTorch 2.4.1 framework. To expedite convergence, the Adam optimizer was employed with exponential decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.999$ for the first and second moment estimates, and a weight decay coefficient of 5×10^{-4}. The learning rate followed a polynomial decay schedule, starting at 0.002 with a total of 80 epochs. For image preprocessing, all datasets were cropped to 256×256 pixels. Considering GPU memory and training stability, the batch size was set to 16 for all datasets. Several data augmentation techniques, including random rotations, flips, and multiscale resizing, were applied during training to improve the model's generalization capability. All experiments were conducted using a fixed random seed of 42 to ensure reproducibility. During evaluation, the patch size used was the same as during training, with an evaluation batch size of 8 for all datasets. To ensure fairness and comparability, all methods were tested under the same experimental conditions.

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Training/validation samples are generated via offline cropping. SAR/optical images and pixel-wise labels are jointly cropped into fixed-size patches during preprocessing, which are used for training and evaluation. A regular grid is used with 256×256 patches, a stride of 256, and zero overlap. Training directly uses the precropped patches with synchronized augmentations (random scaling and random cropping) to improve robustness to scale changes and local perturbations. Since inputs are fixed-size patches, no online sliding-window cropping is performed in training or inference, and online stride/overlap settings do not apply. The 256×256 size is chosen to balance contextual coverage and computational cost: smaller patches reduce mid-scale context, while larger patches increase memory demand and slow training. These settings mainly affect redundancy, context range, and compute overhead, without altering the proposed modeling mechanism. The settings are kept identical across all baselines and ablations for fair and reproducible comparison.

2) Evaluation Metrics: Three sets of evaluation metrics are utilized in this study. The first set focuses on accuracy-related metrics, including overall accuracy (OA), mean intersection over union (mIoU), and Cohen's kappa coefficient (κ, denoted as Kappa in tables). The second set evaluates model complexity, encompassing floating point operations (FLOPs), model parameters (Params), and inference speed (frames per second, FPS). The third set consists of boundary-aware metrics, including boundary intersection-over-union (bIoU) and boundary $F1$-score (Boundary-$F1$). These indicators emphasize the model's ability to preserve object boundaries and capture fine-grained geometric structures—particularly crucial in optical-SAR fusion scenarios where boundary delineation is often blurred or misaligned.

C. Experimental Results and Analyses

To assess the multimodal semantic segmentation performance of the proposed MSFNet, a set of representative baseline methods is selected for comparative analysis. The early CNN-based PSCNN [47] and its improved version DDP [48] serve as baseline networks to evaluate the fundamental abilities of the model in modality fusion and semantic discrimination. Recent multimodal deep networks, including MCANet [34], FTransUNet [49], CFNet [50], CCEnd-Net [51], and ASANet [52], are chosen as highly representative competitors since they incorporate attention modeling, modality guidance, or multiscale feature extraction strategies based on various mechanisms. In addition, several comparison algorithms have introduced interactive fusion strategies, such as SOFANet [36], JoiTriNet-d [53], and DDHRNet [54], which provide a basis for further evaluating MSFNet's performance in structural alignment and fusion consistency. For ablation studies, MSFNet_OPT and MSFNet_SAR denote single-modal variants of the proposed model, which are designed to assess the effectiveness of the dual-modal collaboration mechanism. The comprehensive performance comparison between MSFNet and the aforementioned methods ensures that the evaluation results are both broadly representative and rigorous, with the optimal performance highlighted in red.

1) Experimental Comparison on WHU-OPT-SAR Dataset: Table III provides a quantitative comparison of 11 representative methods on the official WHU-OPT-SAR test set, and Fig. 5 illustrates the visual results for selected regions. As summarized in Table III, MSFNet achieves the best overall balance among the compared methods on the WHU-OPT-SAR dataset. In particular, the proposed method provides stronger OA and mIoU while maintaining more robust class-wise performance on challenging categories such as roads and mixed urban–rural regions. These results indicate that the proposed graph-guided spatial-frequency fusion strategy is effective for integrating complementary optical texture and SAR structural information.

The proposed MSFNet consistently outperforms all comparison models across the three comprehensive evaluation metrics. Compared with the second-best model, MSFNet achieves a 0.19% improvement in OA, a 0.27% increase in Kappa, and a 0.07% gain in mIoU. For the per-class accuracy (UA) across the eight semantic categories, MSFNet particularly enhances the difficult-to-segment classes of farmland and roads by 0.35% and 0.99%, respectively, while maintaining competitive parity with the best baseline for city, village, and water categories. The qualitative comparisons shown in Fig. 5 further validate these statistical findings. In scenarios involving elongated roads and linear water bodies, MSFNet maintains edge continuity by leveraging the spatial-frequency complementary modeling of GSIS-GFIS, effectively avoiding the common breaks and holes observed in existing methods. In urban-farmland mixed regions, the proposed SAGC module effectively exploits SAR scattering features and mitigates the large-scale misclassifications frequently observed in networks such as CaFE and PSCNN. Overall, MSFNet demonstrates stable generalization across multiple land-cover categories and diverse scenes, while ensuring a lightweight design and providing robust fusion of optical textures with SAR geometric information.

2) Experimental Comparison on DFC23 Dataset: Table IV presents the quantitative evaluation results of different semantic segmentation methods on the DFC23 test set, and Fig. 4 provides visual comparisons for selected regions. As summarized in Table IV, MSFNet achieves the best overall performance on the DFC23 dataset, surpassing the strongest baseline in OA, Kappa, and mIoU. Despite the highly fine-grained rooftop categories and strong interclass similarity, the proposed method still shows more stable discrimination across complex roof types. This suggests that the joint use of structure-aware modality alignment and frequency-domain detail enhancement improves semantic differentiation in challenging urban rooftop scenes.

The proposed MSFNet surpasses the existing best method, MCANet, achieving an OA of 89.87%, a Kappa coefficient of 72.15%, and an mIoU of 33.89%, while MCANet obtains 89.53%, 70.88%, and 32.64%, respectively. The mIoU improvement reaches 1.25 percentage points (pp). For the per-class accuracy across the 13 semantic categories, MSFNet achieves the highest UA in five categories, with notable improvements of approximately 1.64% and 2.03% over the second-best method for elongated buildings (Do, 80.51%) and

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small-scale man-made structures (AR, 50.35%), respectively. The qualitative results in Fig. 4 further validate these statistical findings. Traditional CNN-based and transformer-based methods still suffer from road discontinuities and miss small targets in regions with strong scattering noise. By contrast, MSFNet, leveraging GSIS for modality alignment and the Fourier-based GFIS for restoring high-frequency details, produces segmentation maps that closely resemble the GT in terms of building continuity and small target integrity. These results confirm the effectiveness and generalization capacity of the proposed spatial-frequency implicit fusion and dynamic adaptive graph convolution in complex multimodal remote sensing environments.

3) Experimental Comparison on YYX-OPT-SAR Dataset: For the YYX-OPT-SAR dataset, the proposed method demonstrates superior performance compared with the current best model, MCANet. Overall, Table II shows that the proposed model yields consistent gains over strong recent baselines and maintains a clear overall advantage on this challenging optical–SAR benchmark.

As shown in Table II, the overall OA improves from 84.62% to 85.52%, the Kappa coefficient increases from 70.96% to 72.83%, and the mIoU rises from 41.42% to 43.44%. A category-wise comparison indicates notable improvements in bareland, buildings, roads, and background. In particular, the road category improves by 7.22 pp, while the forest class achieves an increase of 11.75 pp, effectively mitigating the problem of missed detections in forested areas. When compared with recent approaches such as DDHRNet, CFNet, and PSCNN, the proposed model consistently achieves an average mIoU improvement of 2–5 pp, maintaining a clear overall lead. The quantitative advantages are further corroborated by the qualitative results in Fig. 6. In typical scenarios such as intersecting roads and complex forest textures, the proposed method improves road connectivity, enhances canopy boundary clarity, and effectively suppresses speckle noise and aliasing artifacts commonly observed in other methods. These results confirm that the graph-guided spatial-frequency fusion strategy effectively integrates complementary information from optical and SAR modalities, yielding robust and precise semantic segmentation performance in complex remote sensing environments.

4) Experimental Comparison on DFC2025 Dataset: On DFC2025, MSFNet achieves an OA of 83.51%, a Kappa of 80.12%, and an mIoU of 70.61%, delivering the best overall performance among all compared methods (Table V). Overall, Table V indicates that MSFNet maintains the best overall performance on DFC2025 while showing clear advantages over both representative multimodal competitors and the single-modal variants.

Compared with DDHRNet, MSFNet improves mIoU by 0.31 pp and OA by 0.25 pp, indicating consistent gains on this challenging all-weather benchmark. Ablation results further verify the benefit of fusion: MSFNet improves mIoU from 63.47% (MSFNet_OPT, using only optical inputs) and 39.64% (MSFNet_SAR, using only SAR inputs) to 70.61%, demonstrating that jointly exploiting optical structural cues and SAR backscatter information is critical for robust land-cover delineation. MSFNet also attains the best performance on the Water class (97.01%, class-wise IoU) and remains competitive on boundary-sensitive categories such as road (76.82%), suggesting improved boundary continuity and reduced speckle-induced fragmentation. Qualitative results in Fig. 7 further support these observations.

D. Boundary-Oriented Evaluation

To explicitly evaluate structural fidelity, we report Boundary-$F1$ and bIoU under tolerance radii $r = \{1, 2, 3, 5, 7\}$. As illustrated in Fig. 8, DDHRNet and FTransUNet achieve the highest Boundary-$F1$, while PSCNN, DDHRNet, and FTransUNet lead in bIoU. Although not the absolute top performer, the proposed MSFNet remains competitive with these state-of-the-art methods and consistently surpasses representative baselines such as DDP, MCANet, and CCEnd-Net. In particular, under stricter tolerance radii ($r = 1, 2$), MSFNet demonstrates strong competitiveness, highlighting the benefits of its structural priors and continuous-domain representation. Combined with the results in Section IV-C, where MSFNet achieves the best OA, Kappa, and mIoU, these findings confirm that MSFNet achieves a favorable balance between overall segmentation accuracy and boundary preservation.

E. Ablation Study on the YYX-OPT-SAR Dataset

To clarify the contribution of the major components in MSFNet, a progressive ablation study was conducted on the YYX-OPT-SAR dataset. Five configurations were evaluated: CNN only, CNN + DA-GCN, CNN + DA-GCN + GSIS, CNN + DA-GCN + GFIS, and the full MSFNet. The evaluation metrics include OA, mIoU, and Kappa, together with model size (MB), Params (M), and FLOPs (G). The results are reported in Table VI.

As shown in Table VI, the full MSFNet achieves the best overall performance, yielding 85.52% OA, 43.44% mIoU, and 72.83% Kappa. Compared with the CNN-only baseline, the full model improves OA by 4.51 points, mIoU by 1.97 points, and Kappa by 6.00 points. In addition, it also reduces model size, parameter count, and FLOPs, indicating that the proposed design improves segmentation accuracy without increasing the overall computational burden.

The intermediate variants do not consistently outperform the CNN-only baseline. This indicates that DA-GCN, GSIS, and GFIS should not be viewed as independently additive modules, but as cooperative components in a unified framework. Specifically, DA-GCN enhances structure- and scattering-aware modeling for SAR data, GSIS performs structure-aware alignment in the continuous spatial domain, and GFIS provides frequency-domain detail compensation and global-context enhancement. Since GSIS and GFIS are coupled through a base-residual design, partial activation of the pipeline cannot fully exploit the intended spatial-frequency complementarity. Consequently, the best performance is achieved only when all three components are jointly enabled.

This observation is consistent with the design in Section III. GSIS generates a structure-aligned spatial base feature, whereas GFIS supplements the missing high-frequency and

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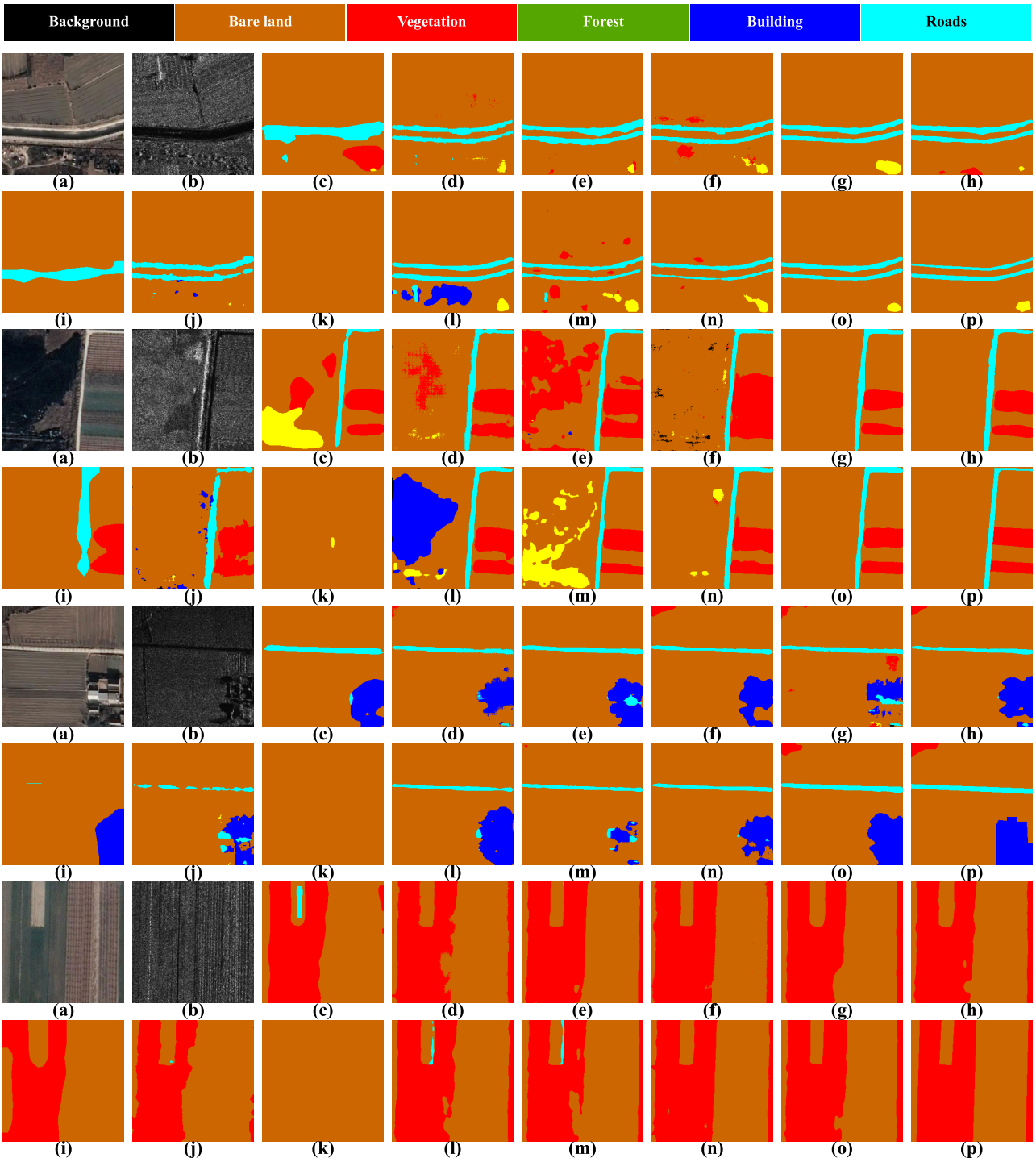


Fig. 6. Qualitative semantic segmentation results on the YYX-OPT-SAR dataset. The subfigures are arranged in the following order: (a) optical image, (b) SAR image, (c) DDP, (d) CFNet, (e) FTransUNet, (f) PSCNN, (g) MCANet, (h) SOFANet, (i) JoiTriNet, (j) DDHRNet, (k) SAR-only, (l) Opt-only, (m) CCEnd, (n) ASANet, (o) MSFNet, and (p) GT. The proposed method produces more coherent region predictions and sharper object boundaries, especially for roads, buildings, and vegetation under cross-modal appearance differences. Class names: background, bareland, vegetation, forest, building, and roads.

global-context information after spatial sampling. Therefore, the gain of the full model is attributed to coordinated spatial-frequency interaction, rather than to any single component alone.

For completeness, additional ablation and variant results are provided in Appendix (Table VII). These results are consistent with the observations in Table VI.

F. Computational Complexity and Efficiency Analysis

The computational complexity and inference efficiency of different deep learning methods on the YYX-OPT-SAR dataset are summarized in Table VIII. To facilitate a more structured comparison of both computational efficiency and segmentation performance, we report model size (MB), parameter count

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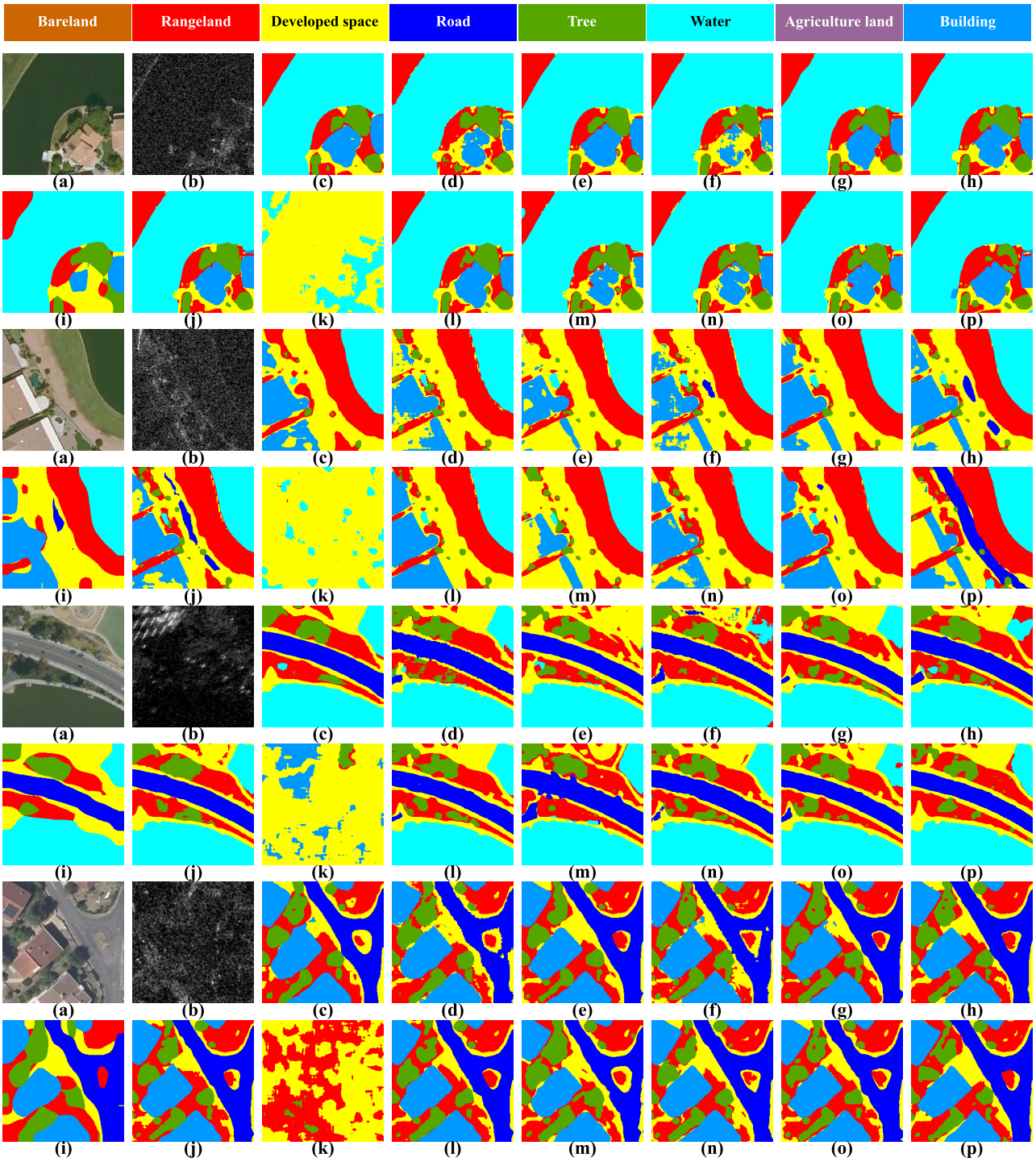


Fig. 7. Qualitative semantic segmentation results on the DFC2025 dataset. The subfigures are arranged in the following order: (a) optical image, (b) SAR image, (c) DDP, (d) CFNet, (e) FTransUNet, (f) PSCNN, (g) MCANet, (h) SOFANet, (i) JoiTriNet, (j) DDHRNet, (k) SAR-only, (l) Opt-only, (m) CCEnd, (n) ASANet, (o) MSFNet, and (p) GT. The proposed method achieves better delineation of roads, buildings, and developed areas while preserving spatial continuity in heterogeneous all-weather scenes. Class names: bareland, rangeland, developed space, road, tree, water, agricultural land, and building.

(M), FLOPs (G), latency (ms), throughput (patch/s), peak GPU memory usage (MB), and mIoU (%). All results are obtained under identical hardware and implementation settings with the same input resolution and batch size, ensuring a fair comparison across methods.

As shown in Table VIII, MSFNet attains 87.08-M parameters and 17.76-G FLOPs, while exhibiting a higher measured latency (192.43 ms/patch) and lower throughput (5.20 patch/s) than lightweight CNN baselines such as PSCNN and CFNet. This discrepancy indicates that wall-clock runtime is not solely determined by FLOPs, and is strongly affected by operator characteristics, including frequency-domain FFT/IFFT and graph-related adjacency/message-passing operations, as well as their memory-access patterns. At the same time, MSFNet

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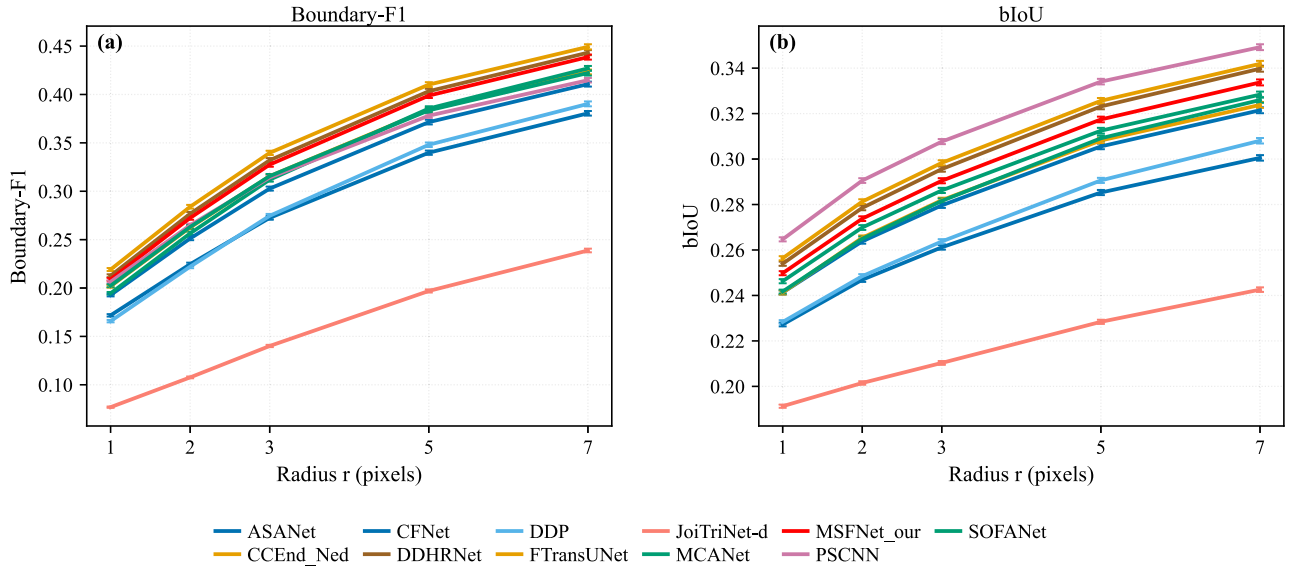


Fig. 8. Boundary-$F1$ and bIoU curves under different tolerance radii on the WHU-OPT-SAR dataset. (a) Boundary-$F1$ and (b) bIoU.

TABLE X
ROBUSTNESS ANALYSIS UNDER NOISE PERTURBATIONS

Noise Level	Metrics	CCEnd-Net	ASANet	MCANet	PSCNN	DDP	DDHRNet	FTransUNet	SOFANet	JoiTriNet-d	CFNet	MSFNet_OPT	MSFNet_SAR	MSFNet
Clean	OA	83.19	84.40	84.62	82.27	80.24	80.78	82.56	82.62	79.91	81.66	82.46	63.59	85.52
	mIoU	47.84	40.79	41.42	39.91	35.76	36.02	39.98	39.18	33.28	38.50	37.97	7.96	43.44
Low	OA	83.10	83.73	72.21	82.16	71.36	8.11	79.56	64.78	41.49	68.32	82.39	62.86	84.18
	mIoU	38.19	39.86	30.17	39.92	24.55	2.06	33.58	10.10	9.67	24.13	38.11	7.90	42.87
Medium	OA	80.98	80.14	70.30	75.03	53.39	6.31	76.79	63.62	9.53	66.27	81.08	63.53	76.85
	mIoU	36.55	37.19	28.58	34.88	11.35	1.01	27.96	8.00	1.51	21.08	37.83	7.97	38.90
High	OA	75.89	77.21	67.44	72.11	19.80	6.02	74.02	63.60	5.88	59.71	73.34	35.78	60.43
	mIoU	33.81	33.86	26.20	30.03	3.43	0.87	23.75	7.95	0.73	17.53	34.62	5.74	32.39

Notes: Noise is injected only at the test stage. For optical images, additive Gaussian noise with $\sigma = 0.01$, 0.03, and 0.05 is applied for Low, Medium, and High levels, respectively. For SAR images, multiplicative speckle noise with equivalent number of looks $L \in \{8, 4, 2\}$ is used, corresponding to the low, medium, and high noise settings, respectively.

also achieves strong segmentation accuracy, reaching 43.44% mIoU, which highlights the tradeoff between performance gain and deployment efficiency.

Compared with lightweight designs, the proposed framework introduces additional computational overhead through the integration of GSIS-GFIS spatial-frequency implicit operators and SAGC-based graph modeling. Although these components improve structural representation and cross-modal alignment, they also increase inference latency and reduce throughput in the current implementation. Therefore, MSFNet should be regarded as a high-precision, high-latency framework rather than an efficiency-oriented model. In its present form, it is more suitable for accuracy-oriented remote sensing analysis than for real-time applications or large-scale rapid-response mapping without substantial optimization.

Future work will focus on improving deployment efficiency through lightweight architectural redesign, more efficient graph construction and message passing, and streamlined spatial-frequency interaction mechanisms.

G. Robustness Analysis

This section evaluates the sensitivity of YYX-OPT-SAR segmentation to controlled residual cross-modal shifts through pixel-level translation. Specifically, the SAR image is shifted by $(\Delta x, \Delta y) = (k, 0)$ pixels, with $k \in \{0, 2, 5, 8\}$, while the optical image and labels remain fixed. Reflect padding is used only to preserve the original input size and maintain a consistent evaluation protocol across methods. Therefore, this experiment should be regarded as a simplified sensitivity analysis under residual pixel-level offsets, rather than a physically complete simulation of real registration errors.

Performance is evaluated in terms of OA, mIoU, and mBF, where mBF is computed with a boundary band of 2 px and a matching tolerance of 2 px. Under perfect alignment, MSFNet attains 85.52% OA, 43.44% mIoU, and 25.00% mBF. For small shifts of 2 and 5 px, the changes remain marginal, indicating robustness to small residual misregistration (Table IX). When the shift increases to 8 px, the metrics decrease slightly, suggesting that large misalignment can still impair boundary

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and region consistency. Overall, MSFNet exhibits graceful degradation as misalignment increases. Noise robustness is assessed by perturbing optical images with additive Gaussian noise, using $\sigma \in \{0.01, 0.03, 0.05\}$, and corrupting SAR images with multiplicative speckle noise, using $L \in \{8, 4, 2\}$. As noise intensifies, MSFNet degrades gradually from the clean setting, maintaining comparatively stable OA and mIoU across low, medium, and high noise levels (Table X). In contrast, the baselines deteriorate more severely under medium and high noise, often falling to single-digit mIoU, whereas MSFNet remains substantially more resilient, demonstrating stronger cross-modal robustness (Table X).

V. Conclusion

MSFNet proposed in this study leverages a dynamic adaptive graph convolutional backbone (DA-GCN) by integrating the structural modeling capabilities of graph neural networks with the continuous mapping advantages of INR, and achieves optical and SAR semantic segmentation. The network employs SAGC in DA-GCN to dynamically encode the scattering geometry of SAR, aligns optical and SAR structures in the continuous coordinate domain using GSIS, and compensates for high-frequency spectral details with GFIS. The dual guidance in the spatial and frequency domains enables MSFNet to maintain fine-grained optical textures while mitigating SAR interpolation distortions, significantly improving boundary continuity and the recovery of structural details. The experimental results demonstrate that MSFNet outperforms existing methods across multiple datasets, particularly in boundary preservation and the recovery of structural details.

Although MSFNet achieves strong segmentation performance, the complexity analysis indicates that its current implementation still incurs considerable computational overhead. In this sense, MSFNet should be regarded as a high-precision, high-latency framework rather than an efficiency-oriented model. Therefore, it is not yet suitable for real-time applications or large-scale rapid-response mapping without substantial optimization. In addition, radar viewing-geometry effects such as layover and shadowing are not explicitly modeled in the current framework. These effects are strongly topography-dependent, as they arise from the interaction between radar line-of-sight geometry and terrain relief, and their impacts under varying imaging conditions have not been systematically evaluated. Future work will focus on improving deployment efficiency and adaptability through the integration of lightweight components, more efficient graph mechanisms, and streamlined spatial-frequency interaction designs, while also exploring the incorporation of DEMs into the graph construction process to better characterize terrain-dependent geometric distortions.

Declaration of Competing Interest

No potential conflict of interest was reported by the authors.

Appendix

To further assess the role of GCN in the INR branch, we replace the GCN-based implicit mapping with a point-wise MLP. The w/o GC variant achieves 80.09% OA, 42.26% mIoU, and 65.65% Kappa, compared with 85.52%, 43.44%, and 72.83% for the full model (Table VII). This result indicates that removing GCN does not cause model failure but leads to consistent performance degradation. Therefore, the role of GCN is better understood as providing a structure-aware inductive bias for modeling neighborhood relationships and improving the robustness of the complete framework.

Overall, the additional ablation results suggest that the proposed components make complementary contributions within MSFNet, while the full model provides the best overall tradeoff.

As shown in Table VII, removing DA-GCN, GSIS, or GFIS degrades performance to different extents. The GCN-related variant remains functional after replacing the graph-based implicit mapping with a point-wise MLP, but still performs worse than the full model, further supporting the role of GCN in enhancing structure-aware neighborhood modeling and overall robustness.

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